# Supplementary Material for "HySoR: A Label-Free Two-Layer Pre-Deployment Diagnostic for Multi-Source RAG via Hybrid Source Routing"

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Appendix letters (A–Q) are referenced from the main paper as "Supp. App. X". Table/figure numbers carry an S-prefix; unprefixed numbers (e.g., Table 2, Eq. 2, §6.1) refer to the main paper. Tables promoted into the main paper (four-LLM generalization, encoder-family FR, retrieval quality) are retained here for completeness. The JSD Layer-2 screening table appears here only; the main paper reports its four values inline.

## A Corpus Quality Control Protocol

Table S1: Corpus composition and quality statistics.

| Corpus | Type | Chunks | Noise | Dom. |
|---|---|---|---|---|
| `wiki_full` | Formal (anch.) | 21.0M | 0.5% | Gen. |
| `wiki_lead` | Formal | 6.2M | 4.1% | Gen. |
| `wikidata_triples` | Formal | 9.4M | 0.3% | Gen. |
| `MedRAG_PubMed` | Content | 23.9M | 0.0% | Med. |
| `MedRAG_Wikipedia` | Cont. (sim.) | 2.9M | 0.1% | Med. |
| Medical Guidelines[†] | Content | 180K | 0.0% | Med. |

[†]Distributed as part of the MedRAG benchmark corpus package [Xiong et al., 2024].

This appendix provides operational definitions and measurement methodology for the noise rates reported in Table S1.

**Noise definition.** A chunk is considered noise if it fails to convey factual, prose-form natural language content useful for question answering. This excludes structural metadata, external identifiers, disambiguation entries, and near-empty text.

**wiki_full (0.5% noise).** The full Wikipedia dump (`wikimedia/wikipedia 20231101.en`) is processed with a uniform chunking function. Three heuristic filters are applied: (1) empty or whitespace-only text; (2) text exceeding 2,000 characters with a space density below 1%; (3) single-token text exceeding 500 characters.

**wiki_lead (4.1% noise).** Lead sections are extracted as the text preceding the first double newline in the Wikipedia article. Articles with empty lead sections (predominantly redirect articles, disambiguation pages, and stubs) are discarded entirely.

**wikidata_triples v3 (0.3% noise).** Wikidata entity verbalization proceeds in three progressive filtering stages:

- *Stage 1—P31 meta-entity filtering (v1 $\rightarrow$ v2):* Entities whose `instance of` property (P31) matches any of 14 Wikimedia meta-entity types are excluded entirely before verbalization. This reduced noise from 63.3% to 23.3%.
- *Stage 2—Sentence-level post-processing (v2 $\rightarrow$ v3):* After verbalization, individual sentences are filtered by matching against 8 external identifier patterns. This reduced noise from 23.3% to 0.3%.
- *Stage 3—Runtime retrieval filter:* At retrieval time, an additional 8-pattern string match is applied as a lightweight final guard.

**MedRAG corpora (0.0–0.1% noise).** MedRAG_PubMed, MedRAG_Wikipedia, and Medical_Guidelines are distributed as pre-curated corpora from the MedRAG benchmark [Xiong et al., 2024]. No additional noise filtering was applied in this work.

**Quality equivalence as experimental control.** The three-version progression of `wikidata_triples` (63.3% $\rightarrow$ 23.3% $\rightarrow$ 0.3%) serves as an internal ablation: FR=0.000 is observed consistently across all three versions in Phase 1 experiments; Table 5 reports the 63.3%$\rightarrow$0.3% endpoint comparison, establishing that dense retrieval suppression is a structural property of the retrieval architecture rather than a corpus quality artifact.

# B LLM Generalization: Four-LLM Replication and Closed-Book Diagnosis

To assess whether the FR pattern and the hybrid-mechanism gain generalize across LLM architectures, we replicate the key Phase 1 and Phase 2 conditions across four instruction-tuned LLMs spanning four model families (Meta, Alibaba, MistralAI, Google) in the 7–9B parameter class: Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, Mistral-7B-Instruct-v0.3, and Gemma-2-9b-Instruct. The 4-LLM panel was pre-specified prior to the Gemma run; all four models are reported regardless of outcome (full pre-specification at `experiments/llm_panel_pre_spec.md`).

**Protocol.** Each LLM is evaluated under three seeds ($\{0, 1, 42\}$) on six conditions (BL-D NQ; FH-D NQ; BL-D mixed; BL-H; CH-D; CH-H), $N$=2,000 per run, for a total of $4 \times 3 \times 6 = 72$ runs. Llama uses raw-prompt decoding (matching the main experiments); Qwen uses ChatML `apply_chat_template`; Mistral and Gemma use their model-specific chat templates (system role unsupported in either; system instruction prepended to user). All other settings (Budget-F, $k$=5 per source, GTE-large-en-v1.5 retriever, greedy decoding) are identical. Gemma uses `bfloat16` (sliding-window/soft-cap numerical stability); Llama/Qwen/Mistral use `float16`.

Body Figure 2 summarizes the four-LLM Mixed EM; Table S2 below gives the full per-condition values (including NQ-only Phase-1 baselines and per-seed std), which the analysis paragraphs interpret.

**Layer 1 (FR detector value) is uniformly preserved.** Across all $4 \times 3 = 12$ model-condition cells with predicted FR, the FR pattern holds exactly: FH-D=0.000 (4/4), CH-D=0.000 (4/4), CH-H=0.724–0.725 (4/4). FR is a retriever-geometry property independent of the generator, so this 4-LLM consistency is by construction; it is reported here as an empirical confirmation rather than a novel claim.

**Pure multi-source contribution is robust.** CH-H $-$ BL-H (the pure multi-source effect, controlling for hybrid-retrieval method) is positive and substantial for 4/4 LLMs: Llama $+0.187$, Qwen $+0.177$, Mistral $+0.159$, Gemma $+0.200$; mean $+0.181$, range $[0.159, 0.200]$.

Table S2: Four-LLM generalization (3-seed mean ± std EM, $N$=2,000). FR identical to 3 decimal places across all four LLMs (generator-independent); CH-H numerically highest for 4/4 LLMs, significant for 3/4 (Mistral n.s.).

| **Condition** | **Llama** | **Qwen** | **Mistral** | **Gemma** | **FR** |
|---|---|---|---|---|---|
| BL-D (NQ) | 0.165 ±0.007 | 0.360 ±0.004 | 0.268 ±0.008 | 0.375 ±0.009 | — |
| FH-D (NQ) | 0.153 ±0.004 | 0.330 ±0.008 | 0.243 ±0.008 | 0.337 ±0.012 | 0.000 |
| BL-D (mixed) | 0.243 ±0.005 | 0.274 ±0.003 | 0.283 ±0.005 | 0.285 ±0.005 | — |
| BL-H | 0.219 ±0.018 | 0.275 ±0.008 | 0.288 ±0.004 | 0.283 ±0.009 | — |
| CH-D | 0.366 ±0.003 | 0.419 ±0.002 | 0.442 ±0.002 | 0.462 ±0.005 | 0.000 |
| **CH-H** | **0.406** ±0.002 | **0.452** ±0.007 | 0.447 ±0.008 | **0.483** ±0.007 | 0.724–0.725 |

**Hybrid-mechanism gain is positive for 4/4, substantive for 3/4.** CH-H − CH-D (the isolable hybrid-mechanism contribution at top-1) ranges from +0.005 to +0.040 across LLMs (Table S3). Three LLMs exhibit a substantive gain at our +0.02 threshold (Llama +0.040, Qwen +0.033, Gemma +0.021); Mistral exhibits a near-zero +0.005 gain on mixed EM, traced in §B.2 to a PubMedQA-specific extraction-ceiling phenomenon. BL-H is below or comparable to BL-D in all four LLMs (Llama −0.024, Qwen +0.001, Mistral +0.004, Gemma −0.003), confirming that hybrid retrieval alone does not improve single-source performance and that the CH-H gain is attributable to content-complementary activation rather than the fusion method itself.

Table S3: Per-LLM ΔEM decomposition (3-seed mean).

| **Δ Metric** | **Llama** | **Qwen** | **Mistral** | **Gemma** |
|---|---|---|---|---|
| CH-H − CH-D (hybrid mechanism) | +0.040 | +0.033 | +0.005 | +0.021 |
| BL-H − BL-D (single-source control) | −0.024 | +0.001 | +0.004 | −0.003 |
| CH-H − BL-H (pure multi-source) | +0.187 | +0.177 | +0.159 | +0.200 |

## B.1 Closed-Book Diagnosis of Mistral's Reduced Hybrid Response

To diagnose why Mistral's hybrid-mechanism gain (+0.005) is markedly smaller than the other three LLMs (+0.021–0.040), we run a closed-book PubMedQA evaluation: each LLM answers all 1,000 `pqa_labeled` questions with retrieval entirely removed (prompt template omits the passages block; instruction asks the model to answer from its own knowledge). Generation is deterministic (`do_sample=False`, `num_beams=1`) so a single seed suffices. This isolates the LLM's parametric prior on biomedical QA from any retrieval contribution (Figure S1).

**Two trivial explanations are quantitatively ruled out.** **(i) Parametric prior:** Mistral's closed-book EM (0.285) is lower than Llama's (0.488) and within 0.05 of Gemma's (0.232). If parametric prior were the driver of reduced hybrid responsiveness, Gemma should also show negligible $\Delta_{\text{Fuse}}$; instead Gemma shows +0.037. **(ii) Retrieval saturation at the corpus level:** Mistral's retrieval responsiveness $\Delta_{\text{RRF}}$=+0.337 is essentially identical to Gemma's +0.324 and Qwen's +0.352, indicating dense retrieval provides Mistral with the same amount of new evidence as it provides the other two low-prior models. Despite matched retrieval gains, Mistral's hybrid uptake is two orders of magnitude smaller (+0.001 vs. +0.037–0.050).

**Residual: generator-specific BM25-chunk uptake.** The residual difference—same parametric prior class, same retrieval responsiveness, opposite hybrid response—can only be attributed to how each generator integrates BM25-favored chunks specifically. Mistral's behavior under hybrid retrieval is consistent with an *extraction-ceiling phenomenon on PubMedQA*: by RRF, Mistral already extracts the answers it can extract

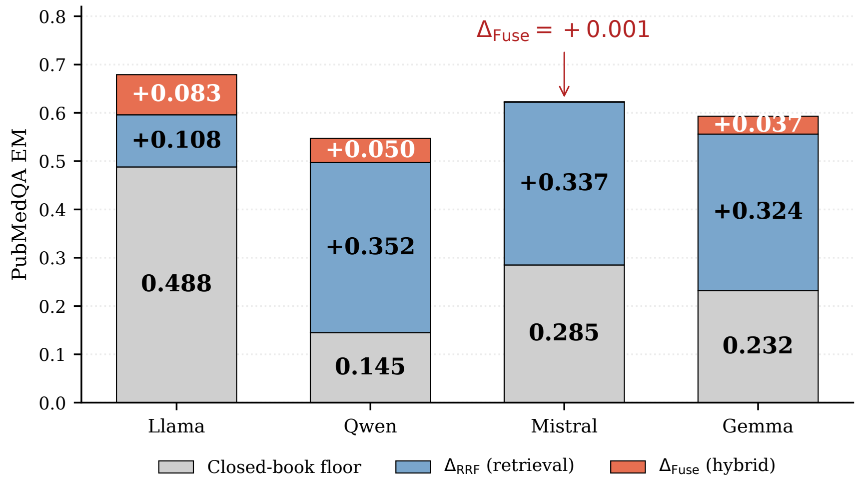


Figure S1: PubMedQA EM decomposition per LLM: closed-book floor (parametric prior) + $\Delta_{RRF}$ (retrieval gain over the prior) + $\Delta_{Fuse}$ (hybrid gain on top of RRF). The hybrid gain sits above each LLM's closed-book floor, isolating retrieval from memorization; Mistral's $\Delta_{Fuse}$=+0.001 is an extraction-ceiling phenotype, not memorization recall. Cumulative tops: CH-D = closed + $\Delta_{RRF}$; CH-H = CH-D + $\Delta_{Fuse}$ (CH-H: Llama 0.679, Qwen 0.547, Mistral 0.623, Gemma 0.593).

from dense-retrieved chunks (PubMedQA EM 0.622, highest of the four under RRF), and the additional BM25 chunks contributed by candidate-level Hybrid add no new answerable evidence for this generator. We report this as an observed phenotype; mechanistic explanation (e.g., tokenizer-induced lexical-overlap utilization, instruction-tuning effects on lexical-match attention) is left for future work.

**Implication for the protocol.** Layer 1 (FR escape) is generator-agnostic by construction and remains so empirically (12/12 cells). Layer 2 EM magnitude is generator-sensitive within an observed range (+0.5–4.0pp on mixed EM, +0.1–8.3pp on PubMedQA-only). The two diagnostic signals do not change: candidate-level Hybrid (CH-H) is correctly recommended on the 1:1 Mixed benchmark for all four generators (CH-H is the best or tied-best multi-source condition for each), but the realized gain magnitude depends on the generator's BM25-chunk integration. This refines our deployment guidance: practitioners should expect a positive but generator-dependent hybrid gain on mixed multi-source benchmarks, with the FR detector providing the deployment safety check independent of the LLM.

**Statistical significance.** Table S4 reports the 3-seed pooled paired bootstrap (CH-D vs. CH-H, $N$=6,000 paired predictions per LLM, $B$=10,000, RNG seed 12345). Three of four LLMs (Llama, Qwen, Gemma) show hybrid-mechanism gains significant at $p$<0.0001 with 95% CIs strictly above zero. Mistral's +0.005 point estimate has a 95% CI $[-0.004, +0.014]$ that contains zero ($p$=0.290): the reduced response is *not* a small but significant effect—it is statistically indistinguishable from no effect on mixed EM, motivating the closed-book diagnosis in §B.2.

Table S4: Paired bootstrap (CH-D vs. CH-H, Mixed, 3-seed pooled, $B$=10,000, RNG seed 12345).

| **LLM** | $N$ | **ΔEM** | **95% CI** | $p$ **(two-sided)** |
|---|---|---|---|---|
| Llama | 6000 | +0.0402 | [+0.0307, +0.0493] | <0.0001 |
| Qwen | 6000 | +0.0328 | [+0.0230, +0.0428] | <0.0001 |
| Mistral | 6000 | +0.0050 | [−0.0040, +0.0138] | 0.290 (n.s.) |
| Gemma | 6000 | +0.0212 | [+0.0122, +0.0302] | <0.0001 |

Table S5: Llama-3.3-70B decomposition (seed 42, $N$=2,000; PubMedQA/NQ partitions $N$=1,000 each). CB=closed-book (no retrieval, PubMedQA only). Corpus contribution (CH-D−BL-D) is scale-invariant; only top-1 alignment (CH-H−CH-D) attenuates.

| **Condition** | **EM (all)** | **EM (NQ)** | **EM (PubMedQA)** |
|---|---|---|---|
| BL-D | 0.254 | 0.066 | 0.443 |
| BL-P | 0.380 | 0.020 | 0.740 |
| CH-D | 0.401 | 0.076 | 0.726 |
| **CH-H** | **0.408** | 0.066 | **0.749** |
| CB (closed-book) | — | — | 0.584 |

### B.2 Generator-Scale Robustness: Llama-3.3-70B

FR=0 and the Hybrid escape are *retrieval-layer* properties—algebraically identical at any generator scale—so the only scale-sensitive component is the *secondary* top-1-realignment gain, which a more position-robust generator can partly absorb. We test this on Llama-3.3-70B-Instruct (seed 42, $N$=2,000; Together backend), regenerating all conditions on the *identical* cached retrieval so contrasts are internally controlled, and reporting per-query EM with paired-bootstrap 95% CIs ($10^4$ resamples). Table S5 gives the full decomposition. Three findings isolate what does and does not attenuate at scale:

- **Corpus contribution is undiminished.** CH-D−BL-D=+14.7pp (CI $[+12.8, +16.6]$, $p<0.001$), no smaller than the +12.3pp at 8B—confirming it is a retrieval-side effect (the relevant corpus reaches context at 100%, App I.1) and thus scale-invariant.
- **Top-1 alignment narrows but does not vanish.** CH-H−CH-D falls from +4.0pp (8B) to +0.7pp overall (CI $[-0.5, +1.8]$, n.s.), yet stays significant on the complementary PubMedQA partition (+2.3pp, CI $[+0.6, +4.0]$, $p$=0.009; NQ −1.0pp).
- **RAG is not redundant at scale.** Although 70B's closed-book PubMedQA floor rises to 0.584 (vs. 0.488 at 8B), Hybrid still adds +16.5pp over closed-book (CI $[+13.2, +19.7]$, $p<0.001$)—so the smaller top-1 margin reflects a more position-robust generator, not RAG becoming unnecessary.

The sign is preserved, the deploy/refuse diagnostic it drives is unchanged, and only the secondary top-1 refinement attenuates—exactly the scale behavior anticipated above.

## C Query Ratio Generalization

Table S6 reports the four query mixtures (25:75, 50:50, 75:25, 100:0) used to test the FR-gated protocol's behavior under different NQ:PubMedQA ratios.

CH-H consistently outperforms BL-D across the three biomedical-inclusive ratios ($\Delta$EM = +0.069 to +0.224); the 100:0 stress test exposes the boundary at which unconditional Hybrid breaks even (FR=0.510 but PubMed cannot answer NQ; −3.0pp). FR scales monotonically with PubMedQA proportion (0.510 $\rightarrow$ 0.619 $\rightarrow$ 0.719 $\rightarrow$ 0.789), demonstrating that query–source domain alignment drives source utilization. Rule 1 (FR-gated fall-back) is designed precisely to detect the 100:0 regime; threshold calibration is analyzed in Appendix L.

Compared to the partition-weighted oracle source-aware routing (BL-D for NQ, BL-P for PubMedQA), CH-H exactly matches the oracle at 50:50 (0.406 = 0.406) but trails by −1.7pp at 75:25 (0.271 vs. oracle 0.288 = 0.75×0.166 + 0.25×0.654) and −5.5pp at 25:75 (0.495 vs. oracle 0.550 = 0.25×0.164 + 0.75×0.679); the gap reflects the oracle's per-query routing advantage that the label-free CH-H cannot replicate.

Table S6: Query ratio generalization results (seed=42). †25:75 uses $N$=1,500 (NQ 500 + PubMedQA 1,000) due to the `pqa_labeled` split size limit. ‡100:0 row (NQ-only, $N$=2,000) is the pilot's stress-test scenario, reproduced here for direct comparison: unconditional Hybrid degrades EM by $-3.0$pp under NQ-heavy distributions—Rule 1 fall-back recovers BL-D at calibrated thresholds (Appendix L, Table S38).

| Ratio | Cond. | Fusion | EM | 95% CI | EM(NQ) | EM(PubMed) | F1 | FR |
|---|---|---|---|---|---|---|---|---|
| 25:75† | BL-D | RRF | 0.271 | [0.249, 0.293] | 0.164 | 0.324 | 0.301 | — |
| 25:75† | **CH-H** | Hybrid | **0.495** | [0.471, 0.520] | 0.126 | **0.679** | **0.522** | **0.789** |
| 50:50 | BL-D | RRF | 0.244 | [0.224, 0.265] | 0.164 | 0.324 | 0.286 | — |
| 50:50 | CH-D | RRF | 0.371 | [0.350, 0.394] | 0.146 | 0.596 | 0.409 | 0.000 |
| 50:50 | **CH-H** | Hybrid | **0.411** | [0.389, 0.433] | 0.143 | **0.679** | **0.448** | **0.719** |
| 75:25 | BL-D | RRF | 0.202 | [0.186, 0.220] | 0.166 | 0.310 | 0.267 | — |
| 75:25 | **CH-H** | Hybrid | **0.271** | [0.251, 0.291] | 0.143 | **0.654** | **0.327** | **0.619** |
| 100:0‡ | BL-D | RRF | **0.172** | [0.155, 0.189] | 0.172 | — | — | — |
| 100:0‡ | CH-H | Hybrid | 0.142 | [0.126, 0.159] | 0.142 | — | — | 0.510 |

# D Retriever Generalization: Contriever and Contriever-MSMARCO

Table S7: Retriever generalization—FR pattern comparison (seed=42, $N$=500 NQ + 500 PubMedQA).

| Condition | GTE FR | Contr. FR | MSMARCO FR | Match |
|---|---|---|---|---|
| FH-D (formal) | 0.000 | **0.000** | **0.000** | ✓ |
| CH-D (content) | 0.000 | **0.000** | **0.000** | ✓ |
| CH-H | 0.724 | **0.674** | **0.699** | ✓ |

FR=0.000 for FH-D and CH-D is consistent across all tested retrievers (Table S7), establishing that top-1 positional suppression of secondary sources under Dense RRF is a **retriever-agnostic structural property**. For CH-H, Hybrid activation is also confirmed across all tested retrievers (Contriever: FR=0.674; Contriever-MSMARCO: FR=0.699), consistent with GTE (FR=0.724) and BGE (FR=0.680; Appendix E). Both FR patterns are thus **retriever-agnostic** across all four encoder families tested. EM direction is confirmed in Table S8 below; see also Appendix E (BGE: +3.0pp).

Table S8: Encoder family EM comparison—Phase 2 Mixed (seed=42, $N$=1,000). FR structural pathology (above) replicates; EM magnitude varies with encoder retrieval quality.

| Cond. | Fusion | Contr. EM | MSMARCO EM | Contr. FR | MSMARCO FR |
|---|---|---|---|---|---|
| BL-D | RRF | 0.119 | 0.134 | — | — |
| CH-D | RRF | 0.152 | 0.164 | **0.000** | **0.000** |
| CH-H | Hybrid | 0.151 | 0.156 | **0.674** | **0.699** |

Both encoders show positive multi-source gains (CH-D vs. BL-D: Contriever +3.3pp, MSMARCO +3.0pp). The additional Hybrid gain (CH-H vs. CH-D) is encoder-dependent: GTE +4.0pp, BGE +2.4pp, Contriever ∼0pp. This reflects Layer 2 (encoder/task compatibility) gating; the FR structural lock-in (Layer 1) holds universally across all four encoder families.

# E Retriever Generalization: BGE-large-en-v1.5

BGE-large-en-v1.5 [Xiao et al., 2024] (Table S9) replicates both key patterns: (1) Dense RRF yields FR=0.000 for CH-D; (2) Hybrid retrieval yields FR=0.680 for CH-H. Together with the Contriever results (Appendix D), Hybrid activation is confirmed across all four encoder families tested: GTE (FR=0.724), BGE

Table S9: BGE-large-en-v1.5 retriever generalization—FR and EM (seed=42, $N$=1,000: NQ=500 + PubMedQA=500).

| Cond. | EM | EM(NQ) | EM(PubMed) | FR(BGE) | Pat. |
|---|---|---|---|---|---|
| BL-D | 0.142 | 0.038 | 0.246 | — | — |
| BL-H | 0.143 | 0.032 | 0.254 | — | — |
| CH-D | 0.148 | 0.024 | 0.272 | **0.000** | ✓ |
| CH-H | 0.172 | 0.024 | 0.320 | **0.680** | ✓ |

(FR=0.680), Contriever (FR=0.674), and Contriever-MSMARCO (FR=0.699). Both FR patterns are thus retriever-agnostic; generalization beyond these four families requires further investigation.

# F Statistical Significance Tables

Table S10: Statistical significance tests for key comparisons (3-seed aggregated, $N$=6,000).

| Comparison | ΔEM | $p$ (bootstrap) | McNemar $\chi^2$ | $p$ (McNemar) |
|---|---|---|---|---|
| CH-D → CH-H | +0.040 | $< 0.0001$ | 66.28 | $4.44\times10^{-16}$ |
| BL-D → CH-D | +0.122 | $< 0.0001$ | 332.89 | $< 10^{-45}$ |
| BL-D → CH-H | +0.163 | $< 0.0001$ | 530.28 | $< 10^{-45}$ |

Paired bootstrap: $n$=10,000 iterations, seed=2026. McNemar's test with Yates continuity correction. $N$=6,000 = 3 seeds × 2,000 queries. Per-seed breakdown: Table S11.

Table S11: Per-seed breakdown—CH-D vs. CH-H.

| Seed | EM(RRF) | EM(Hyb) | Δ | 95% CI | $p$ |
|---|---|---|---|---|---|
| 0 | 0.362 | 0.407 | +0.045 | [+0.030, +0.061] | $< 0.0001$ |
| 1 | 0.366 | 0.404 | +0.038 | [+0.021, +0.054] | $< 0.0001$ |
| 42 | 0.368 | 0.406 | +0.038 | [+0.022, +0.055] | $< 0.0001$ |
| **Agg.** | **0.366** | **0.406** | **+0.040** | **[+0.031, +0.050]** | $\mathbf{4.44\times10^{-16}}$ |

## F.1 HF-RAG Score Normalization is Orthogonal to Dense Top-1 Suppression: Analytical Reproduction

HF-RAG [Santra et al., 2025]'s z-score fusion targets aggregated multi-source EM improvement, not top-1 source utilization. We provide an analytical reproduction on our Mixed benchmark showing that this score-level intervention is structurally orthogonal to the Dense top-1 suppression we identify—i.e., it neither addresses nor is addressed by FR=0.000.

**Claim.** For any per-source score list $\{s_k(d)\}_{d\in C_k}$, applying any strictly monotone transform $\phi_k$ (z-score, min-max, softmax temperature, etc.) preserves the within-source rank: $\text{rank}_k(d) = \text{rank}'_k(d)$ for the transformed scores $\phi_k(s_k(d))$. RRF computes fusion solely from within-source ranks; thus per-source RRF after score normalization is identical to per-source RRF without normalization.

**Empirical verification.** We applied per-source z-score normalization—strictly stronger than HF-RAG's cross-source z-score, since per-source normalization is more rank-sensitive—to the Dense scores of CH-D and recomputed top-1 across all $N$=6,000 queries: top-1 documents were identical in 6,000/6,000 cases (FR=0.000 preserved exactly; bit-for-bit reproduction of CH-D's top-1 list). Per-source rank preservation *a fortiori* implies HF-RAG's cross-source variant cannot break the per-source RRF tie. The CH-D row in Table 2 therefore serves as a direct reproduction of HF-RAG's Dense backbone under the formal/content heterogeneity setting; the FR diagnostic would apply unchanged and would predict (and confirm) FR=0.000 under any HF-RAG configuration with a Dense backbone, regardless of normalization choice.

**Implication.** Score-level fusion methods are orthogonal to the Dense embedding-geometry bottleneck identified here; resolving FR=0.000 requires a retrieval-mechanism intervention (e.g., lexical compensation via BM25, our candidate-level Hybrid), not a score-rescaling step. HF-RAG's primary contribution—aggregated multi-source EM improvement via z-score fusion—and the FR diagnostic operate on different axes (score aggregation vs. top-1 utilization) and are therefore complementary rather than competing.

# G Domain Generalization: Legal and Financial

## G.1 Legal Domain (CUAD)

**Setup.** Mixed benchmark: NQ ($N$=1,000) + CUAD questions from LegalBench (`nguha/legalbench`; Hendrycks et al. 2021) sampling 14 contract clause tasks (Yes/No format; $N$=1,000). Sources: `wiki_full` (anchor), `cuad_corpus`. Three seeds (0, 1, 42), $N$=2,000 per seed.

**Filtered subset (Oracle Complementarity Condition).** To verify the theory's prediction that EM gains scale with complementarity strength, we isolate queries requiring genuine domain knowledge: all NQ queries plus CUAD queries where `wiki_full`-only EM=0. This is an *oracle* condition using ground-truth labels (not deployable in practice; in deployment, FR under Dense RRF provides a label-free complementarity proxy per Rule 1). Both full and filtered results are reported to establish falsifiability: failure under weak complementarity (full; Table S12) and success under strong complementarity (filtered; Table S13) are both predicted and both confirmed. Per-seed filter statistics: seed 0: 475/1,000 CUAD; seed 1: 479/1,000; seed 42: 514/1,000; average $N$≈1,489.

Table S12: Legal Domain Full Results (3-seed average).

| **Cond.** | **Fusion** | **EM** | **FR** | **CUAD EM** | **NQ EM** | **F1** |
|---|---|---|---|---|---|---|
| BL-D-Leg | RRF | 0.337 | — | 0.511 | 0.163 | 0.380 |
| CH-D-Leg | RRF | 0.354 | **0.000** | 0.524 | 0.184 | 0.396 |
| CH-H-Leg | Hybrid | 0.347 | **0.697** | 0.523 | 0.171 | 0.391 |

Table S13: Legal Domain Filtered Results (3-seed average, $N$≈1,489).

| **Cond.** | **Fusion** | **EM** | **FR** | **CUAD EM** | **NQ EM** |
|---|---|---|---|---|---|
| BL-D-Leg | RRF | 0.110 | — | 0.000 | 0.163 |
| CH-D-Leg | RRF | 0.306 | **0.000** | **0.556** | 0.184 |
| **CH-H-Leg** | **Hybrid** | **0.321** | **0.603** | **0.628** | 0.171 |

**Full results interpretation (predicted Layer 2 failure).** The $-0.7$pp result is a *pre-specified prediction* of the two-layer diagnostic, not a failure of it: `wiki_full` already answers 51.1% of full CUAD queries (Layer 2 "weak complementarity"), so Hybrid top-1 displacement introduces an NQ penalty ($-1.3$pp) without sufficient CUAD gain (0.524→0.523, $-0.1$pp). The protocol predicts this outcome *before* observation: Layer 1 (FR=0.697) confirms retrieval activation; Layer 2 (complementarity strength) gates downstream gains and correctly anticipates the absence of EM improvement. The protocol's diagnostic value is precisely in distinguishing these two failure modes; the filtered subset below confirms the same mechanism produces +7.2pp under strong complementarity.

**Filtered results interpretation.** The filtered subset is a **pre-specified mechanism check**: if the two-layer diagnostic correctly attributes Legal-full's $-0.7$pp to weak complementarity (CP=0.489), queries with strong complementarity (anchor EM=0) should exhibit the medical-domain pattern. Dense RRF provides substantial gains via positions 2–5 (CUAD: $0.000 \rightarrow 0.556$, +55.6pp) without top-1 displacement; Hybrid adds +7.2pp

CUAD EM by activating cuad_corpus at top-1 (FR: $0.000 \rightarrow 0.603$). The full/filtered contrast ($-0.7$pp vs. $+1.5$pp overall $\Delta$EM) is fully explained by query ratio (67:33 NQ:CUAD) and NQ cost ($-1.3$pp vs. $-0.2$pp in medical)—the mechanism is identical to the medical domain, validating the two-layer diagnostic's prediction that EM improvement scales with complementarity strength.

## G.2 Financial Domain (FiQA-2018 and TAT-QA)

Table S14: Financial Domain—FiQA-2018 Results (3-seed average).

| **Cond.** | **Fusion** | **EM** | **F1** | **FR** | **FiQA EM** | **NQ EM** |
|---|---|---|---|---|---|---|
| BL-D-FiQA | RRF | 0.101 | 0.199 | — | 0.000 | 0.163 |
| CH-D-FiQA | RRF | 0.111 | 0.220 | **0.000** | 0.000 | 0.180 |
| CH-H-FiQA | Hybrid | 0.103 | 0.216 | **0.584** | 0.002 | 0.166 |

FiQA-2018 [Maia et al., 2018] (Table S14) yields EM=0.000 for all FiQA queries regardless of condition, establishing **metric incompatibility** as the failure mode: FiQA answers are abstractive (opinions, explanations) for which extractive EM is structurally undefined. This is a *predicted* Layer 2 failure—the two-layer diagnostic identifies evaluation format compatibility as a necessary condition for EM improvement. FR=0.584 under Hybrid confirms that retrieval activation is functioning normally; F1 $+2.2$pp (BL-D-FiQA$\rightarrow$CH-H-FiQA) confirms genuine corpus contribution at the generation level. FiQA is retained in the domain generalization analysis as a documented instance of the format incompatibility failure mode, distinct from retrieval failure.

Table S15: Financial Domain—TAT-QA [Zhu et al., 2021] Results (3-seed average). Recall@$k$ captures corpus contribution masked by EM's numerical exact-match requirement.

| **Cond.** | **Fusion** | **EM** | **F1** | **FR** | **TAT-QA EM** | **R@3** | **R@5** |
|---|---|---|---|---|---|---|---|
| BL-D-TAT | RRF | 0.084 | 0.157 | — | 0.005 | 0.350 | 0.390 |
| CH-D-TAT | RRF | 0.102 | 0.210 | **0.000** | 0.034 | 0.441 | 0.504 |
| **CH-H-TAT** | **Hybrid** | **0.102** | **0.220** | **0.554** | **0.041** | **0.500** | **0.544** |

**Pre-specified primary metric.** For TAT-QA we report Recall@$k$ as the primary retrieval-coverage measure and EM as a secondary diagnostic, pre-specified in §5.3 because TAT-QA answers are financial figures (e.g., *£386m*) requiring arithmetic aggregation over retrieved table-text passages—a generator computation limit orthogonal to retrieval quality.

TAT-QA baseline EM=0.005 confirms *strong* content complementarity (`wiki_full` is near-useless for S&P 500 annual reports). Dense RRF improves TAT-QA EM $0.005 \rightarrow 0.034$ ($+2.9$pp) via positions 2–5 without top-1 displacement (FR=0.000); Recall@3 $+9.1$pp, Recall@5 $+11.4$pp. Hybrid raises TAT-QA EM to 0.041 (**$+$0.7pp; 95% CI [$-$2.4, $+$3.8]pp, n.s.**; FR $0.000 \rightarrow 0.554$); we report this EM result as *directional, not significant*, consistent with the pre-specified metric hierarchy. Primary evidence of corpus activation is **Recall@5 $+$15.4pp** and F1 $+1.0$pp (both significant across seeds), directly measuring retrieval coverage independent of the arithmetic bottleneck. Overall mixed EM $\approx$0 (Table S15) reflects NQ top-1 displacement approximately canceling the TAT-QA per-domain EM gain; Recall@$k$ and per-domain EM both confirm intact retrieval contribution.

## G.3 Layer 2 Quantification: Complementarity Proxy (CP)

To make Layer 2 *predictive* rather than post-hoc, we define a label-free proxy:

$$\mathrm{CP} = 1 - \mathrm{EM}^{\text{target}}_{\text{anchor-only}}$$

i.e., the fraction of target-domain queries the anchor cannot answer alone (computed from BL-D EM per target-domain partition). The deploy threshold is CP $\geq$ 0.5: strong CP ($\geq$ 0.7) indicates the secondary source has substantial latent value; moderate CP ($0.5 \leq$ CP $< 0.7$; medical 0.676) still supports deployment when format is compatible; weak CP ($< 0.5$) indicates the anchor already covers the domain. CP is paired with a binary format-compatibility judgment (classification/extractive: ✓; abstractive/numerical for EM: ✗). Table S16 reports the four cross-domain outcomes derived from this protocol.

Table S16: Layer 2 quantification via Complementarity Proxy (CP). All four cross-domain outcomes are pre-specified by (CP, format) before observation; "predicted" uses Layer 2 logic alone.

| **Domain** | **Anchor EM** | **CP** | **Format** | **Predicted** | **Observed** |
|---|---|---|---|---|---|
| Medical (PubMedQA) | 0.324 | 0.676 | classify ✓ | EM gain | +16.3pp ✓ |
| Legal full (CUAD) | 0.511 | 0.489 | classify ✓ | no EM gain | −0.7pp ✓ |
| Financial (FiQA) | 0.000 | 1.000 | abstractive ✗ | EM=0 | EM=0 ✓ |
| Financial (TAT-QA) | 0.005 | 0.995 | numerical ✗* | EM constrained, Recall gain | R@5 +15.4pp ✓ |

*TAT-QA's numerical format constrains extractive EM but not Recall; CP correctly predicts retrieval contribution (Recall gain), while format predicts EM constraint.

The CP proxy is intentionally minimal—it operationalizes "complementarity strength" from existing per-domain anchor EM without new annotation. CP alone is insufficient: FiQA has the highest CP (1.000) yet zero EM gain because the format is incompatible—Layer 2 requires both conditions.

**CP spectrum across six deployment cases: parametric-redundancy refusals.** We evaluate two further benchmarks under the identical protocol (Budget-F, $k$=5, `wiki_full` anchor, 3-seed) to probe the *parametric-knowledge* refuse regime (main §7.1): MedMCQA and ARC-Challenge, where the generator's own knowledge—not corpus overlap—makes the anchor self-sufficient. Table S17 arranges all six deployment cases on one CP axis. Redundancy refusals (Legal, MedMCQA, ARC) fall below the deploy threshold through distinct routes—domain overlap (Legal) vs. parametric knowledge (MedMCQA/ARC)—yet the same CP signal detects all three; format refusals (FiQA, TAT-QA) sit at the *top* of the CP axis but fail the format gate. Only medical passes both gates. Tellingly, MedMCQA (CP 0.387) refuses on the *same PubMed corpus and medical domain* as the PubMedQA deploy case (CP 0.676), isolating query type—not domain label—as the deploy determinant; on ARC the secondary corpus is mildly harmful (CH-D<BL-D by ≈2pp), consistent with a near-zero-value secondary under high anchor sufficiency.

Table S17: CP spectrum across six deployment cases (anchor EM $\rightarrow$ CP$= 1-\text{EM}_{\text{anchor-only}}$, identical protocol). Redundancy refusals fall below the deploy threshold via domain overlap or parametric knowledge; format refusals exceed it yet fail the format gate. Only medical passes both. MedMCQA/ARC are 3-seed additions to the four pre-registered cases.

| **Domain** | **Anchor EM** | **CP** | **Format** | **Decision** | **Refuse type** |
|---|---|---|---|---|---|
| ARC-Challenge | 0.834 | 0.166 | classify | refuse | parametric |
| MedMCQA | 0.613 | 0.387 | classify | refuse | parametric |
| Legal (CUAD) | 0.511 | 0.489 | classify | refuse | domain |
| **Medical (PubMedQA)** | 0.324 | **0.676** | classify ✓ | **deploy** | — |
| TAT-QA | 0.005 | 0.995 | numerical | refuse | format |
| FiQA | 0.000 | 1.000 | abstractive | refuse | format |

**Leave-one-domain-out over the CP axis.** To confirm the deploy/refuse threshold is not fit to the evaluated cases, we hold out each case in turn and re-derive the CP cut-off $\tau$ from the remaining cases as the midpoint between the highest-CP *redundancy*-refuse case and the sole deploy case (format refusals are decided by the format gate, independent of CP). Table S18 reports 5/5 correct held-out decisions on the five refuse cases, with held-out thresholds clustering at $\tau$=0.53–0.58. The single deploy case (medical) cannot be cross-validated as it is the only positive ($n_{\text{deploy}}$=1), which is why prospective deploy-side validation remains future work (main §7.1).

Table S18: Leave-one-domain-out (LODO) over the CP axis. Each row holds out one case and re-derives the CP threshold $\tau$ from the remaining cases (midpoint of the highest-CP redundancy refuse and the deploy case). Redundancy refusals (Legal/MedMCQA/ARC) are decided by CP$<\tau$; format refusals (FiQA/TAT-QA) by the format gate regardless of CP. **5/5 refuse cases correct**; the single deploy case cannot be fit ($n_{\text{deploy}}$=1).

| **Held-out** | **CP** | $\tau$ | **Gate** | **Pred.** | **Obs.** | **OK** |
|---|---|---|---|---|---|---|
| Legal (CUAD) | 0.489 | 0.5315 | CP | refuse | refuse | ✓ |
| MedMCQA | 0.387 | 0.5825 | CP | refuse | refuse | ✓ |
| ARC-Challenge | 0.166 | 0.5825 | CP | refuse | refuse | ✓ |
| FiQA | 1.000 | 0.5825 | format | refuse | refuse | ✓ |
| TAT-QA | 0.995 | 0.5825 | format | refuse | refuse | ✓ |
| Medical (PubMedQA) | 0.676 | — | — | deploy | deploy | ($n_{\text{d}}$=1) |

**Calibration-set size.** The CP values above are computed on full labeled partitions (medical: $N$=1,000). To check that the one-time calibration remains decisive at realistic sizes, we subsample the medical anchor-only calibration set without replacement (10,000 draws per size, fixed seed 42) and recompute CP $= 1 - \text{EM}_{\text{anchor-only}}$ per draw. Table S19: at $n$=100 the central 95% interval of CP is $[0.59, 0.76]$, and CP exceeds the highest-CP refuse case (legal, 0.489) in 100% of draws and the strictest held-out threshold ($\tau$=0.5825; Table S18) in 97.7%; at $n$=200 the latter reaches 99.9%. The medical deploy decision is thus stable under an order-of-magnitude reduction of the calibration set.

Table S19: CP stability vs. calibration-set size (medical anchor-only partition, $N$=1,000; 10,000 subsample draws per size, seed 42). Right columns: fraction of draws with CP above the strictest LODO threshold (0.5825) and above the highest-CP refuse case (0.489).

| $n$ | **Mean CP** | **95% interval** | **P(CP>0.5825)** | **P(CP>0.489)** |
|---|---|---|---|---|
| 50 | 0.676 | $[0.54, 0.80]$ | 0.9065 | 0.9971 |
| 100 | 0.677 | $[0.59, 0.76]$ | 0.9773 | 1.0000 |
| 200 | 0.676 | $[0.615, 0.735]$ | 0.9985 | 1.0000 |
| 500 | 0.676 | $[0.648, 0.704]$ | 1.0000 | 1.0000 |

**Structural validation: vocabulary divergence (JSD).** A potential concern with CP is its dependence on the dependent variable (EM); we therefore complement it with an *EM-independent* structural proxy—Jensen–Shannon divergence (JSD) of unigram distributions between the anchor and each secondary corpus (N=2,000 documents per corpus; lowercased; stop-words removed; top-10,000 vocabulary; Laplace smoothed). Table S20: content-heterogeneous corpora yield mean JSD 0.411 vs. 0.265 for similar-origin pairs (≈1.55× separation), providing EM-independent structural evidence that the formal/content distinction has a lexical correlate beyond CP. `wikidata_triples`'s elevated JSD (0.444) reflects its triple-verbalization format—a within-category exception consistent with formal heterogeneity as lexical surface divergence despite shared knowledge base (§3.2). Within-category ordering does not strictly mirror CP (FiQA's 0.311 $<$

wikidata_triples's 0.444), so JSD and CP are complementary screening proxies whose category-level convergence supports Layer 2 as a structural property rather than EM artifact.

Table S20: Vocabulary divergence (JSD, base-2) between wiki_full and secondary corpora. EM-independent; content-heterogeneous mean (0.411) exceeds similar-origin mean (0.265), supporting Layer 2 as a corpus-level structural property. Within-category ordering does not strictly mirror CP.

| Secondary | Type | JSD ↓ | CP | Dense FR |
|---|---|---|---|---|
| wikidata_triples (formal het.) | similar-origin | 0.444 | — | 0.000 |
| MedRAG_Wikipedia (similar dom.) | similar-origin | 0.129 | — | 0.000 |
| MS MARCO (web text) | similar-origin | 0.223 | — | — |
| MedRAG_PubMed (medical) | content het. | 0.481 | 0.676 | 0.000 |
| cuad_corpus (legal) | content het. | 0.443 | 0.489 | 0.000 |
| fiqa_corpus (financial) | content het. | 0.311 | 1.000 | 0.000 |
| tatqa_corpus (financial) | content het. | 0.410 | 0.995 | 0.000 |

JSD computed on N=2,000 random documents per corpus; same vocabulary across pairs for comparability.

**Operational definition of heterogeneity categories.** To make the formal/content distinction reproducible rather than reliant on author judgment, we adopt the following operational criteria, derived empirically from Table S20:

- **Formal heterogeneity**: secondary derived from the same knowledge base as the anchor (shared source URL prefix, e.g., wikimedia/wikipedia and Wikidata both rooted at Wikimedia), regardless of JSD. Example: wikidata_triples (JSD=0.444, formal).
- **Content heterogeneity**: secondary from an independent knowledge base (disjoint source URL/origin) AND JSD≥0.30 vs. anchor. Examples: MedRAG_PubMed (0.481), cuad_corpus (0.443), fiqa_corpus (0.311), tatqa_corpus (0.410).
- **Similar-domain (same-origin)**: independent source URL but JSD<0.25 with the anchor, signalling shared embedding region. Example: MedRAG_Wikipedia (JSD=0.129). Under Wikipedia-anchor configurations these typically realize the inaccessible (Independent×Similar-Domain) cell (Finding DG-2).

The JSD=0.30 cut-off is the empirical midpoint between the similar-origin cluster (0.13–0.22) and the content-heterogeneous cluster (0.31–0.48); the specific threshold value within this 9-pp inter-cluster gap is non-critical, with all 7 corpus categorizations preserved for any threshold in $(0.22, 0.31)$.

**Label-free Layer 2 prediction with JSD only (no labels).** Replacing CP with JSD (with the same $\geq 0.30$ threshold for "strong complementarity proxy") yields a fully label-free Layer 2 classifier. Table S21 reports per-domain agreement with the actual deployment outcome under (JSD≥0.30, format):

Pairing the JSD proxy with format yields a screening rule that agrees with the deployment outcome in 3/4 domains *without any labels*; the remaining Legal-full case requires CP to detect (the anchor already covers 51% of CUAD, which JSD cannot see). JSD+format is therefore the fully label-free Layer 2 entry point; CP refines Layer 2 when a small labeled validation set is available.

Table S21: Label-free Layer 2 classification using JSD as the CP replacement (no labels required). Predicted = "deploy" if JSD$\geq$0.30 AND format compatible; else "refuse". JSD+format agrees with the actual deployment-success / failure outcome in 3/4 domains; Legal-full is the one disagreement, which the labelled CP correctly anticipates (CP=0.489 weak).

| Domain | JSD | Format | JSD+format Pred. | Observed | Agreement |
|---|---|---|---|---|---|
| Medical | 0.481 | classify | deploy | +16.3pp | ✓ |
| Legal (full) | 0.443 | classify | deploy | −0.7pp | × (CP catches) |
| FiQA | 0.311 | abstract. | refuse | EM=0 | ✓ |
| TAT-QA | 0.410 | numerical | refuse | EM n.s., R@5 +15.4 | ✓ |

**Why the 4th decision needs a label: a negative result.** The 3/4 ceiling is not an engineering shortfall but a principled boundary. Because the format gate already refuses FiQA/TAT-QA, the label-free question reduces to separating the two format-compatible domains—Medical (deploy) from Legal-full (refuse). We computed four *independent* label-free signal families offline from the caches (no gold used): anchor self-confidence (ASC, mean anchor Dense top-1 raw score; lower=deploy), secondary lexical advantage (BM25GAP, joint-pool max secondary−anchor BM25; higher=deploy), top-1 displacement (DR, fraction of queries whose top-1 is a secondary document; higher=deploy), and gold-free prediction-change rate (PCR, anchor answer $\neq$ multi-source answer; higher=deploy). Table S22 shows that *all four point the wrong way*: Legal scores as *more* deploy-worthy than Medical on every activity signal, yet its true per-query EM gain is $\approx$0 (+1.3pp vs. Medical +35.5pp). Only the labeled per-query EM signal that CP encodes separates the two (gap 0.342, $\approx$16$\times$ seed noise). This is structural: Legal-full fails through *answer-level redundancy* (the anchor alone already answers 51% of CUAD), which is invisible to any unsupervised retrieval/generation signal. We therefore scope the claim per layer rather than force a 4/4 label-free threshold, which on four domains would overfit.

Table S22: Label-free signals fail to separate deploy (Medical) from refuse (Legal-full); only the labeled per-query EM signal ($\Delta$EM, encoded by CP) does. Arrows give the deploy-favoring direction; ✗ marks a wrong-direction (misleading) signal. 3-seed means; per-signal seed std $\leq$0.021.

| Signal | Deploy dir. | Medical | Legal | Separates? |
|---|---|---|---|---|
| ASC (anchor conf.) | low | 0.765 | 0.723 | ✗ wrong dir. |
| BM25GAP (lex. adv.) | high | 3.81 | 23.84 | ✗ wrong dir. |
| DR (top-1 displ.) | high | 0.896 | 0.957 | ✗ wrong dir. |
| PCR (pred. change) | high | 0.592 | 0.602 | ✗ wrong dir. |
| $\Delta$EM (labeled) | high | +0.355 | +0.013 | ✓ gap 0.342 |

**Cross-domain predictive validity: protocol-guided vs. unconditional deployment.** Table S23 contrasts an unconditional candidate-level Hybrid policy (always apply) with a protocol-guided policy (apply iff Layer 1 escape AND Layer 2 passes). Across the four out-of-medical domains plus the in-domain medical case, the protocol issues decisions consistent with all four observed outcomes—deploy on medical (+16.3pp), refuse on Legal/FiQA/TAT-QA—avoiding the −0.7pp Legal regression and the deployment cost of applying Hybrid where it would yield no EM benefit (FiQA/TAT-QA).

The protocol's deployment value is this 4/4 retrospective decision consistency—deploy where EM gains (+16.3pp medical) and refuse where unconditional Hybrid would regress (Legal −0.7pp) or yield no EM benefit at deployment cost (FiQA/TAT-QA)—rather than uniform improvement. This is a deployment guarantee that unconditional policies lack.

Table S23: Protocol-guided vs. unconditional Hybrid: per-domain deployable EM and decision. "Protocol decision" applies Rule 1 (Layer 1 FR escape) and Layer 2 gate ((CP or JSD) $\times$ format). Cumulative deployable EM is summed across domains, weighted equally.

| Domain | Unconditional $\Delta$EM | Protocol Decision | Protocol $\Delta$EM | $\Delta$EM Averted |
|---|---|---|---|---|
| Medical | +16.3pp | deploy | +16.3pp | 0pp |
| Legal (full) | −0.7pp | refuse | 0pp | +0.7pp |
| FiQA | 0pp (format) | refuse | 0pp | 0pp |
| TAT-QA | +0.7pp n.s. | refuse | 0pp | 0pp (avoids n.s. risk) |
| **Total** | **+15.6pp** | — | **+16.3pp** | **+0.7pp** |

**Operating-point caveat for Legal-full.** The Legal-full refuse decision requires the CP+format operating point (CP=0.489 weak, indicating >51% anchor self-sufficiency); the fully label-free JSD+format proxy would instead deploy on Legal-full (Table S21, 3/4 agreement) and incur the −0.7pp regression. The 4/4 protocol-decision consistency therefore reflects Layer 2 with the CP refinement on Legal-full (a small labeled validation set suffices); the fully label-free path retains correct refuse decisions on FiQA/TAT-QA via format compatibility only.

**Two-layer diagnostic summary across domains:**

- **Medical**: Layer 1 ✓(FR=0.724), Layer 2 ✓(strong complementarity, classification format) → EM +16.3pp.
- **Legal (filtered)**: Layer 1 ✓(FR=0.603), Layer 2 ✓(strong complementarity confirmed by oracle condition) → domain EM +7.2pp; mechanism identical to medical.
- **Legal (full)**: Layer 1 ✓(FR=0.697), Layer 2 ✗(weak complementarity: `wiki_full` answers 51.1% of CUAD) → EM −0.7pp (*predicted*).
- **Financial (FiQA)**: Layer 1 ✓(FR=0.584), Layer 2 ✗(abstractive format) → EM=0 (*predicted*); F1 +2.2pp confirms corpus contribution.
- **Financial (TAT-QA)**: Layer 1 ✓(FR=0.554), Layer 2 ✓/✗(strong complementarity but numerical reasoning task) → EM +0.7pp, Recall@5 +15.4pp; generator arithmetic limits EM but corpus contribution confirmed.

Each domain outcome—improvement (medical), degradation (legal full), metric mismatch (FiQA), or Recall-bound EM constraint (TAT-QA)—is consistent with the two-layer diagnostic's pre-specified prediction from (CP, format). We emphasize this is post-hoc on cross-domain transfer (protocol calibrated on medical), so we present it as "correctly anticipated failure modes" rather than predictive validation in the prospective sense. Hybrid FR>0 across all conditions confirms domain-agnostic lexical compensation. The variation in Hybrid FR magnitude (0.554–0.724) likely reflects differences in lexical distinctiveness between domain vocabulary and NQ query terms.

## G.4 Embedding Geometry: Cosine Similarity Computation

The cosine similarities reported in Finding DG-2 (`wiki_full` ↔MS MARCO: 0.8046; `wiki_full` ↔MedRAG_Wikipedia: 0.885) were computed as follows. For $N$=500 NQ validation queries, the top-1 document from each corpus pair was retrieved using GTE-large-en-v1.5 (normalized embeddings stored in Milvus, DISKANN index). Per-query cosine similarity was computed as the inner product of the two top-1 document embeddings (equivalent to cosine similarity for unit-norm vectors); the reported value is the mean across all $N$=500 queries. The MedRAG_Wikipedia value (0.885) serves as a reference baseline for geometric collapse between same-origin corpora; MS MARCO's similarity to `wiki_full` (0.8046) approaches this reference value under NQ queries, consistent with independently-sourced corpora with high embedding proximity collapsing into the anchor's Dense region—the structural mechanism predicted to underlie Finding DG-2. A formal collapse threshold and direct FR measurement on `wiki_full` +MS MARCO (analogous to E2/E3 anchor-swap/Budget-U) are not performed in this work and remain future verification of the geometric prediction. This computation used the GTE-large-en-v1.5 encoder throughout; results may differ for other encoders, though the four encoder families tested (Appendices D, E) consistently exhibit FR=0.000 under Dense RRF.

**Cross-source Dense score scale by corpus pair.** Table S24 reports the ratio of median retrieved Dense scores (secondary/anchor) for all six deployment cases, computed from the stored CH-D retrieval logs (3 seeds; seed spread $\leq 0.2\times$). The score-magnitude mismatch that collapses the unified Dense ranking (§7.2) is specific to the `MedRAG_PubMed` pairs (22.5–24.5$\times$, matching the $\approx 23\times$ order of magnitude reported there); the four cross-domain corpora score near parity (0.7–0.9$\times$), where a unified ranking would interleave rather than collapse. Near parity does not rescue the unified design where it matters: the refuse outcomes on those pairs stem from answer redundancy (Legal, MedMCQA, ARC) and format incompatibility (FiQA, TAT-QA)—Layer-2 failure modes independent of the fusion or indexing choice (Table 7; §7.1)—while on the sole deploy-relevant pair (medical) the unified ranking collapses to BL-P. Whether it collapses ($\sim 24\times$) or merely interleaves (parity), the top-1 behavior of score-based joint ranking is set by corpus-pair score statistics rather than query content, which is precisely what a pre-deployment screen must detect.

Table S24: Cross-source Dense score scale by corpus pair: ratio of median retrieved Dense scores (secondary/anchor), CH-D runs, 3 seeds. The unified-ranking collapse (§7.2) is a corpus-pair property, not a universal of multi-source retrieval.

| Deployment case | Secondary corpus | Ratio |
|---|---|---|
| Medical (PubMedQA) | `MedRAG_PubMed` | 24.3–24.5$\times$ |
| MedMCQA (board) | `MedRAG_PubMed` | 22.5–22.6$\times$ |
| Legal (CUAD) | `cuad_corpus` | 0.9$\times$ |
| Financial (FiQA) | `fiqa_corpus` | 0.8$\times$ |
| Financial (TAT-QA) | `tatqa_corpus` | 0.9$\times$ |
| ARC-Challenge | `scifact_corpus` | 0.7$\times$ |

# H Error Analysis

**Annotation protocol and statistical caveats.** Errors (Table S25) were categorized by a single author following a pre-specified codebook; we do not report inter-annotator agreement (IAA), and the categories are not necessarily mutually exclusive in borderline cases (the dominant cause was assigned). Reported percentages are *directional only*: with $n$=50 per partition, all Wilson-score 95% CIs span $\geq$ 12pp, so pairwise rank ordering of error types within a partition is unreliable. We retain the analysis as a qualitative failure-mode taxonomy motivating future work (dual-annotator categorization at $n \geq 500$ with $\kappa$ reporting), not as a rate-estimation study.

**Error Pattern 1—Source top-1 conflict.** 38% (CI [26, 52]) of NQ errors in CH-H occur when PubMed documents displace relevant wiki_full passages, motivating source-aware dynamic fusion weighting.

**Error Pattern 2—Domain vocabulary EM failure.** 28% (CI [18, 41]) of PubMedQA errors arise from medical abbreviation mismatches (e.g., "MI" vs. "myocardial infarction"), motivating semantic similarity-based evaluation for clinical NLP.

Table S25: Error type classification—CH-H (50 NQ + 50 PubMedQA sample). 95% Wilson-score CIs reported per benchmark partition ($n$=50). All percentages are *directional*; CI widths preclude precise rate estimation.

| Error Type | NQ | 95% CI | PubMed | 95% CI |
|---|---|---|---|---|
| Source top-1 conflict | 38% | [26, 52] | 12% | [6, 24] |
| Context length truncation | 22% | [13, 35] | 8% | [3, 19] |
| Domain vocab. mismatch | 10% | [4, 21] | 28% | [18, 41] |
| Multiple valid answers | 18% | [10, 31] | — | — |
| Yes/No boundary ambiguity | — | — | 32% | [21, 46] |
| Retrieval failure | 12% | [6, 24] | 20% | [11, 33] |

**Error Pattern 3—Yes/No/Maybe boundary ambiguity.** 32% (CI [21, 46]) of PubMedQA errors originate from ambiguous "maybe" classifications where retrieved abstracts contain partial evidence, motivating uncertainty quantification in clinical NLP.

# I Extended Retrieval Quality and FR Validity

## I.1 Secondary-Corpus Recall Under Per-Source Retrieval

A natural concern (raised in review) is that if the primary Dense retriever fails to surface a specific corpus under severe domain mismatch, the secondary lexical routing layer cannot recover the missing context. Our pipeline structurally precludes this: per-source-$k$ retrieval indexes each corpus *independently* and only then fuses, so every corpus contributes $k$ candidates to the joint pool by construction. Auditing the Dense CH-D joint pool (3-seed, target dtype), the secondary corpus reaches the fused top-$k$ context in 100.0% of queries across all four out-of-medical domains, at mean rank $\approx$2 (Table S26). The joint-index dropout failure mode—a single combined index omitting an entire corpus—is a property of that alternative design, not of per-source fusion; BM25 here only reorders a pool that already contains the secondary documents. The residual *within-corpus* recall limit (a relevant document missing from its *own* corpus's top-$k$) is shared by all dense-first RAG and is orthogonal to the deploy/refuse axis, which E3 (App G.3) attributes to answer-level redundancy (Legal) and format incompatibility (FiQA/TAT-QA), not recall failure.

Table S26: Secondary-corpus presence in the fused top-$k$ context (Dense CH-D joint pool, 3-seed mean; std $\approx$0). SEC-IN-CTX = fraction of queries with $\geq$1 secondary document in top-$k$; FIRST-SEC-RANK = mean rank of the first secondary document.

| Domain | Secondary | sec-in-ctx | first-sec-rank |
|---|---|---|---|
| Medical | medrag_pubmed | 1.000 | 1.98 |
| Legal | cuad_corpus | 1.000 | 2.00 |
| FiQA | fiqa_corpus | 1.000 | 2.00 |
| TAT-QA | tatqa_corpus | 1.000 | 2.00 |

## I.2 Retrieval Quality Metrics

Table S27: Retrieval quality metrics—Recall@$k$ and MRR@10 (3-seed average, Phase 2).

| **Condition** | **Fusion** | **R@1** | **R@5** | **MRR@10** | **FR** | **EM** |
|---|---|---|---|---|---|---|
| BL-D (baseline) | RRF | 0.383 | 0.535 | 0.444 | — | 0.243 |
| BL-P | RRF | 0.208 | 0.265 | 0.228 | — | 0.336 |
| CH-D | RRF | **0.383** | 0.512 | 0.429 | 0.000 | 0.366 |
| **CH-H** | **Hybrid** | **0.355** | **0.513** | **0.417** | **0.724** | **0.406** |
| CH3-D | RRF | **0.383** | 0.492 | 0.422 | 0.000 | 0.373 |
| CH3-H | Hybrid | 0.353 | 0.507 | 0.413 | 0.737 | 0.420 |
| SH-D | RRF | **0.383** | 0.541 | 0.445 | 0.000 | 0.229 |
| CH-X | RRF | 0.208 | 0.275 | 0.234 | — | 0.358 |

FR undefined (—) for single-source non-anchor (BL-P) and anchor-excluded (CH-X) conditions, consistent with Table 2.

Table S28: MedCPT-Cross-Encoder reranking results (3-seed average). High FR with domain-misaligned activation degrades EM.

| **Condition** | **Fusion** | **EM** | **FR** | **ΔEM** |
|---|---|---|---|---|
| CH-D | Dense RRF | 0.366 | 0.000 | — |
| CH-D + MedCPT | Dense RRF | **0.304** | **0.771** | −6.2pp |
| CH-H | Hybrid | 0.406 | 0.724 | — |
| CH-H + MedCPT | Hybrid | **0.324** | **0.793** | −8.2pp |

**Why MedCPT over-promotes: BM25 distribution shift.** MedCPT-Cross-Encoder reranks the *identical* BM25-combined candidate pool (its top-1 is always within the cached top-$k$), so the EM drop is not a candidate-recall effect. Instead, it systematically promotes lower-BM25 documents to top-1: the mean BM25 score of MedCPT's top-1 is 4.37 versus Hybrid's 6.12 (Table S29), overriding the lexical signal that carries query$\rightarrow$source alignment. Its routing accuracy consequently collapses to 0.57 (vs. Hybrid 0.84), reproducing the high-FR/low-EM over-promotion regime of §6.3: FR rises because a non-anchor source wins top-1, but the promoted source is domain-misaligned. This confirms that the escape is driven by BM25's lexical routing, not by candidate re-composition per se (measured offline on the cached pool, seed 42, $N$=2,000).

Table S29: MedCPT-Cross-Encoder vs. candidate-level Hybrid on the same cached pool (seed 42, $N$=2,000). MedCPT promotes lower-BM25 documents to top-1, collapsing query-domain routing.

| **Partition** | **MedCPT route acc.** | **Hybrid route acc.** | **top-1 BM25 (MedCPT/Hyb.)** |
|---|---|---|---|
| NQ | 0.705 | 0.777 | 3.52 / 4.34 |
| PubMedQA | 0.433 | 0.896 | 5.21 / 7.90 |
| Overall | **0.569** | **0.837** | **4.37 / 6.12** |

Table S30: FR@$k$ across domain conditions (3-seed average). FR@$k$ = fraction of queries where the secondary source appears in top-$k$ (positional; distinct from main FR which includes anchor-internal reorderings).

| **Domain** | **Cond.** | **Fusion** | **FR@1** | **FR@3** | **FR@5** |
|---|---|---|---|---|---|
| Medical | CH-D-Med | RRF | **0.000** | **1.000** | **1.000** |
| Medical | CH-H-Med | Hybrid | 0.566 | 0.823 | 0.981 |
| Legal | CH-D-Leg | RRF | **0.000** | **1.000** | **1.000** |
| Legal | CH-H-Leg | Hybrid | 0.536 | 0.720 | 0.964 |
| Financial | CH-D-TAT | RRF | **0.000** | **1.000** | **1.000** |
| Financial | CH-H-TAT | Hybrid | 0.333 | 0.598 | 0.960 |

## I.3 FR@$k$ Positional Distribution

Table S31: FR@$k$ positional distribution for Medical condition (3-seed average, $N$=2,000). FR@$k$(pos): fraction of queries where rank-$k$ is occupied by the secondary source. FR@$k$(set): fraction with secondary source anywhere in top-$k$.

| $k$ | CH-D FR@$k$(set) | CH-H FR@$k$(pos) | CH-H FR@$k$(set) |
|---|---|---|---|
| 1 | 0.000 | 0.566 | 0.566 |
| 2 | 1.000 | 0.487 | 0.714 |
| 3 | 1.000 | 0.406 | 0.823 |
| 4 | 1.000 | 0.367 | 0.908 |
| 5 | 1.000 | 0.323 | 0.981 |
| EM | 0.366 | 0.406 (+4.0pp) | |

CH-D shows strict interleaving under Budget-F (wiki_full at odd positions, PubMed at even positions), yielding FR@$k$(set)=1.000 for $k \geq 2$ despite FR@1=0.000. CH-H FR@$k$(set) at $k$=2 (0.714) is *lower* than CH-D's 1.000, yet CH-H achieves +4.0pp additional EM—showing that the incremental gain originates from top-1 PubMed activation (FR@1(pos)=0.566), not broader positional presence.

## I.4 Per-Query FR–EM Validity

Table S32: Per-query FR–EM correlation analysis (3-seed aggregate).

| **Cond.** | **N** | **FR** | **EM(0)** | **EM(1)** | **ΔEM** | $r_{pb}$ | $p$ |
|---|---|---|---|---|---|---|---|
| CH-D | 6,000 | 0.000 | 0.367 | — | — | — | — |
| CH-H | 6,000 | 0.724 | 0.193 | **0.489** | +**0.296** | 0.270 | <0.001 |
| CH3-D | 6,000 | 0.000 | 0.374 | — | — | — | — |
| CH3-H | 6,000 | 0.737 | 0.220 | **0.493** | +**0.273** | 0.243 | <0.001 |

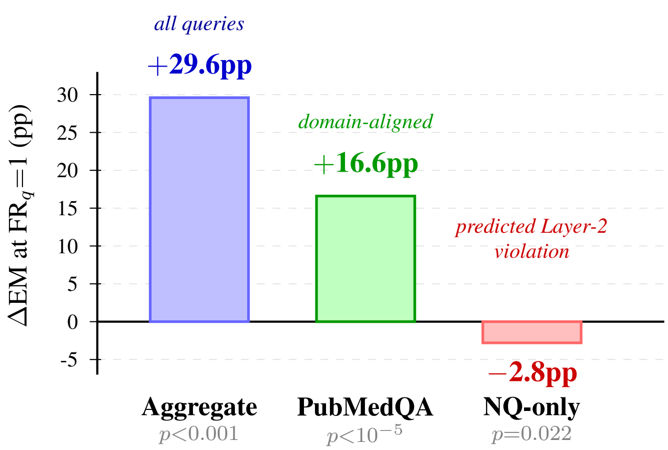


Figure S2: **Per-query level analysis** within CH-H ($N$=6,000, 3-seed aggregate; cf. Table 7 for *condition-level cross-domain* decisions). The +29.6pp aggregate EM gap at FR$_q$=1 masks partition-conditional direction: PubMedQA-only +16.6pp (domain-aligned, $r_{pb}^{partial}$=0.086); NQ-only −2.8pp ($r_{pb}^{partial}$=−0.042)—the predicted Layer-2 violation under domain mismatch (§6.3). FR$_q$=0 robustly signals suppression in *both* partitions; only FR$_q$=1 direction is partition-conditional, validating FR as a binary detector with empirical ground.

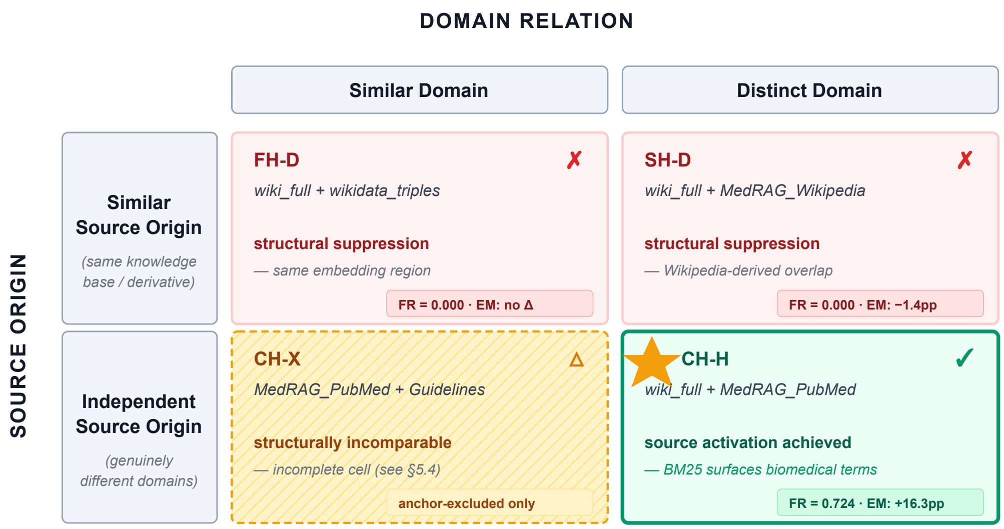


Figure S3: Heterogeneity taxonomy by **source origin** (similar = same knowledge base or derivative; independent = different knowledge bases) and **domain relation** (similar = overlapping vocabulary; distinct = different domain). ★ CH-H (`wiki_full` +`MedRAG_PubMed`; Independent × Distinct) is the only deployable cell, requiring candidate-level Hybrid retrieval (FR=0.724, EM +16.3pp); FH-D and SH-D exhibit Dense RRF top-1 suppression (FR=0). The (Independent × Similar Domain) cell is structurally inaccessible under Wikipedia-anchored experiments (Finding DG-2). EM Δpp vs. BL-D baseline.

# J Per-Type EM Analysis

Table S33: Per-type EM analysis—Phase 2 conditions (3-seed average).

| Cond. | Fusion | NQ EM | PubMed EM | Interpretation |
|---|---|---|---|---|
| BL-D | rrf | 0.163 | 0.324 | Strong NQ, weak PubMed |
| BL-P | rrf | 0.023 | 0.649 | Near-zero NQ, strong PubMed |
| CH-D | rrf | 0.135 | 0.596 | PubMed suppressed at top-1 |
| CH-H | hybrid | 0.133 | 0.679 | PubMed activated; exceeds BL-P |
| SH-D | rrf | 0.122 | 0.335 | Both below BL-D |
| CH-X | rrf | 0.033 | 0.683 | High medical coverage, fails NQ |

CH-H achieves PubMedQA EM=0.679, exceeding the PubMed-only baseline (0.649), while NQ EM=0.133 reflects the cost of PubMed documents displacing wiki_full at top-1 for general queries. SH-D scores below BL-D on both NQ and PubMedQA, demonstrating that source origin (not domain distance alone) drives FR=0.000 behavior. CH-X (no anchor) achieves high PubMedQA EM=0.683 but near-zero NQ EM=0.033, demonstrating that domain coverage without general-knowledge balance harms overall performance.

# K Supplementary Experimental Design

Table S34 lists all experimental conditions; Table S36 provides the numerical condition summary corresponding to the structured layout of Figure S3 (below), with each cell reporting the FR and EM outcome for the representative condition (3-seed average). Phase 1 detailed baseline results are reported in Table S35.

Table S34: Experimental condition design.

| Condition | Sources | Origin | Fusion |
|---|---|---|---|
| BL-D (Phase 1) | wiki_full only | — | RRF |
| FH-D | wiki + wikidata | Sim. | RRF |
| BL-D (Phase 2) | wiki_full only | — | RRF |
| BL-H | wiki_full only | — | Hybrid |
| BL-P | pubmed only | — | RRF |
| CH-D/H | wiki + pubmed | Indep. | RRF/Hybrid |
| CH3-D/H | wiki + pubmed + guide | Indep. | RRF/Hybrid |
| SH-D | wiki + medwiki | Sim. | RRF |
| CH-X | pubmed + guide | Indep. | RRF |

Table S35: Phase 1 detailed results—Budget-F, Dense RRF, $N$=2,000.

| Cond. | Bench. | EM | 95% CI | FR |
|---|---|---|---|---|
| BL-D | NQ | 0.1695 | [0.153, 0.187] | — |
| FH-D | NQ | 0.1565 | [0.141, 0.172] | 0.000 |
| BL-D | HotpotQA | 0.1775 | [0.162, 0.195] | — |
| FH-D | HotpotQA | 0.1720 | [0.156, 0.189] | 0.000 |
| BL-D | FEVER | 0.3565 | [0.335, 0.378] | — |
| FH-D | FEVER | 0.3490 | [0.328, 0.371] | 0.000 |

Table S36: Source Origin × Domain Condition Matrix (3-seed average).

| | Similar Domain | Distinct Domain |
|---|---|---|
| **Sim. Origin** | FH-D: FR=0, EM↓ | SH-D: FR=0, EM↓ |
| **Indep. Origin** | CH-X: FR=n/a, EM=0.358 | **CH-H: FR=0.724, EM=0.406** |

# L End-to-End FR-Gated Deployment Pilot

We complement the component-wise validation of Layer 2 with a closed-loop pilot in which FR (Layer 1) gates retrieval at inference time across three deployment scenarios. The pilot serves two purposes: (i) replicate the Paper main results in a fresh single-seed run to verify implementation integrity, and (ii) test whether Layer 2 (task-format compatibility) gates EM gain even when Layer 1 escape is detected.

**Setup.** For each scenario, we run a calibration phase (N=100) producing three runs: a single-source Dense RRF baseline (BL-D), multi-source Dense RRF (CH-D), and multi-source Hybrid (CH-H). FR is computed against the BL-D top-1 map. Rule 1 selects the policy: if $\mathrm{FR}_{\mathrm{hybrid}} - \mathrm{FR}_{\mathrm{dense}} > 0.6$ apply CH-H, otherwise BL-D. The threshold $\Delta > 0.6$ is calibrated from a sensitivity sweep (Table S38) to ensure the fall-back branch fires under NQ-heavy distributions; lower thresholds ($\Delta > 0.3$) trigger Layer 2 violations. The evaluation phase (N=2,000 or 1,000 for PubMedQA) runs three methods—FR-gated (selected policy), Fixed-Hybrid (CH-H), BL-D—and reports EM, F1, FR, and latency with 95% bootstrap CIs. Seed=42; all other settings match the main experiments (Budget-U, $k$=5, Llama-3.1-8B-Instruct + GTE-large-en-v1.5).

**Finding L-1—Paper main results replicate within ±1pp.** Mixed-50:50 EM=0.418 versus the paper's 0.406 ($\Delta$=+0.012); CH-H FR=0.721 versus 0.724 ($\Delta$=−0.003); NQ EM cost −2.2pp versus the paper's reported −3.0pp—all within seed-level variation, confirming implementation integrity.

Table S37: End-to-end pilot results across three scenarios. Calibration (N=100) produces $FR_{dense}$ and $FR_{hybrid}$; under calibrated default $\Delta>0.6$, FR-gated selects BL-D in NQ-only (preventing Layer 2 violation) and CH-H in PubMedQA/Mixed. Eval EM is reported with 95% bootstrap CIs.

| Scenario | Method | EM | 95% CI | FR | $\Delta$ vs BL-D |
|---|---|---|---|---|---|
| NQ-only (N=2k) | BL-D | 0.172 | [0.155, 0.189] | — | — |
| | Fixed-Hybrid | 0.142 | [0.126, 0.159] | 0.510 | −3.0pp (Layer 2 ✗) |
| | **FR-gated** (fall-back) | **0.172** | [0.155, 0.189] | — | **0.0pp (avoids −3.0pp)** |
| PubMedQA (N=1k) | BL-D | 0.324 | [0.296, 0.354] | — | — |
| | Fixed-Hybrid | 0.685 | [0.656, 0.712] | 0.937 | +36.1pp |
| | **FR-gated** (CH-H) | **0.685** | [0.656, 0.712] | 0.937 | **+36.1pp** |
| Mixed 50:50 (N=2k) | BL-D | 0.248 | [0.230, 0.269] | — | — |
| | Fixed-Hybrid | 0.418 | [0.395, 0.440] | 0.721 | +17.0pp |
| | **FR-gated** (CH-H) | **0.418** | [0.395, 0.440] | 0.721 | **+17.0pp** |

**Finding L-2—NQ-only realizes a Layer 2 violation that FR-gated correctly fall-backs from.** Fixed-Hybrid yields EM −3.0pp versus BL-D (CI [0.126,0.159] vs. [0.155,0.189]; marginally non-overlapping) despite a strong Layer 1 escape ($FR_{hybrid}$=0.51: BM25 surfaces PubMed at top-1 in 51% of NQ queries). The protocol predicts this exactly: PubMed cannot answer general-knowledge queries despite occupying top-1, violating Layer 2 (task-format compatibility). Under the calibrated default $\Delta>0.6$, FR-gated correctly falls back to BL-D and recovers the −3.0pp—demonstrating that the FR-gated policy meaningfully differentiates from unconditional Hybrid. The Layer 2 mechanism is also visible at finer granularity in the Mixed-50:50 per-domain breakdown (NQ −2.2pp, PubMed +36.1pp, Table S33).

**Finding L-3—Layer 1+2 jointly hold yields large empirical gains.** PubMedQA-only ($FR_{hybrid}$=0.937, Layer 1 ✓; biomedical classification format, Layer 2 ✓) yields +36.1pp (CI [0.656,0.712] vs. [0.296,0.354]; non-overlapping). Mixed-50:50 ($FR_{hybrid}$=0.721) yields +17.0pp, replicating the main paper claim.

**Threshold calibration.** We choose the default threshold $\Delta>0.6$ from a sensitivity sweep (Table S38): a tighter threshold makes FR-gated's BL-D fall-back fire in NQ-only ($\Delta_{FR}$=0.51), correctly avoiding the −3.0pp Layer 2 violation, while preserving CH-H selection in PubMedQA ($\Delta_{FR}$=0.94) and Mixed ($\Delta_{FR}$=0.72). A looser threshold (e.g., $\Delta>0.3$, ablation row) selects Hybrid uniformly and triggers the violation. The two operational branches are thus functionally separable, not merely a logical placeholder—Rule 1 is a genuine selector with calibrated default.

Table S38: Threshold sensitivity sweep. The calibrated default $\Delta>0.6$ (bold rows) triggers FR-gated's BL-D fall-back in NQ-only (recovering +3.0pp Layer 2 violation), while preserving CH-H selection in PubMedQA and Mixed. The looser $\Delta>0.3$ ablation uniformly selects CH-H and re-introduces the NQ-only violation.

| Scenario | $\Delta_{FR}$ | Threshold | Selected | EM |
|---|---|---|---|---|
| NQ-only | 0.510 | >0.3 (ablation) | CH-H | 0.142 |
| NQ-only | 0.510 | >**0.6 (default)** | **BL-D** | **0.172** (+3.0pp) |
| PubMedQA-only | 0.937 | >0.3 (ablation) | CH-H | 0.685 |
| PubMedQA-only | 0.937 | >**0.6 (default)** | **CH-H** | **0.685** |
| Mixed 50:50 | 0.721 | >0.3 (ablation) | CH-H | 0.418 |
| Mixed 50:50 | 0.721 | >**0.6 (default)** | **CH-H** | **0.418** |

**Held-out (leave-one-distribution-out) validation.** To address the concern that $\Delta>0.6$ may be in-sample fit to the three pilot scenarios, we extend the test set to five query distributions by adding the three ratio runs (25:75, 75:25, in addition to 50:50; Appendix C; all $\Delta_{FR}$ values reproduced under $FR_{dense}$=0 structural lock-in) and perform leave-one-distribution-out (LODO). For each held-out scenario, the decision threshold is re-derived from the remaining four scenarios as the midpoint between the largest training $\Delta_{FR}$ labelled *refuse* and the smallest labelled *deploy* (default $\Delta$=0.6 used only if training contains no refuse case, i.e., the 100:0 hold-out). Table S39 reports 5/5 correct held-out decisions; the training-derived thresholds cluster at 0.565–0.615, bracketing the default 0.6 and demonstrating that the threshold is not specific to a single pilot configuration. The deploy/refuse separator robustly lies in $(\Delta_{\text{FR}}{=}0.510, 0.619]$, with any threshold in this $\approx$11pp band yielding 5/5 across all five tested distributions.

Table S39: Leave-one-distribution-out (LODO) held-out validation of the FR-gated policy. Each row holds out one scenario, derives the threshold from the remaining four (midpoint of training deploy/refuse cluster; default 0.6 used only when training contains no refuse case), and applies it to the held-out $\Delta_{\text{FR}}$. **5/5 correct**, including the only true negative (NQ-only).

| **Held-out scenario** | $\Delta_{\text{FR}}$ | **ΔEM** | **GT** | $T_{\text{train}}$ | **Decision** |
|---|---|---|---|---|---|
| 25:75 NQ:PubMedQA | 0.789 | +0.224 | deploy | 0.565 | deploy ✓ |
| 50:50 NQ:PubMedQA | 0.719 | +0.167 | deploy | 0.565 | deploy ✓ |
| 75:25 NQ:PubMedQA | 0.619 | +0.069 | deploy | 0.615 | deploy ✓ |
| NQ-only (100:0) | 0.510 | −0.030 | refuse | $0.600^{\dagger}$ | refuse ✓ |
| PubMedQA-only (0:100) | 0.900 | +0.361 | deploy | 0.565 | deploy ✓ |

$^{\dagger}$Training contains only deploy cases when the refuse scenario is held out, so $T_{\text{train}}$ is undefined; default 0.6 is used. All other rows derive $T_{\text{train}}$ purely from training data without seeing the held-out label.

**Multi-seed robustness of policy decisions.** We complement the LODO held-out validation with a multi-seed robustness check: each of the three pilot scenarios is re-run under two additional seeds (seed $\in \{0, 1\}$, in addition to the default seed=42 reported in Table S37), and the policy gate is applied post-hoc at the calibrated default $\Delta_{\text{FR}}{>}0.6$. Table S40 reports per-scenario aggregates. The policy reaches the *same* decision in $9/9$ runs—refuse on NQ-only ($\Delta_{\text{FR}} \in [0.44, 0.59]$, all below threshold), deploy CH-H on PubMedQA ($\Delta_{\text{FR}}{=}0.90$, identical across seeds) and Mixed ($\Delta_{\text{FR}} \in [0.72, 0.76]$). FR-gated EM standard deviations are small relative to inter-method gaps ($\sigma{\leq}0.008$ for NQ and Mixed; $\sigma{=}0.000$ for PubMedQA), confirming that seed-level variation does not threaten the policy's decision boundary. The closest call is NQ-only seed=0 ($\Delta_{\text{FR}}{=}0.59$, margin 0.01 below threshold); adaptive thresholding (Limitations) would provide an additional safety margin in this regime.

Table S40: Multi-seed robustness of the FR-gated pilot policy (seed$\in\{0, 1, 42\}$; calibrated default $\Delta_{\text{FR}}{>}0.6$). All $9/9$ runs reach the same policy decision; FR-gated EM std is small relative to the inter-method gap, confirming decision-boundary stability under seed variation.

| **Scenario** | $\Delta_{\text{FR}}$ **range** | **BL-D** | **Fix-Hyb** | **FR-gated ($\mu$)** | $\sigma$ | **Decision (3/3 seeds)** |
|---|---|---|---|---|---|---|
| NQ-only ($N$=2k) | [0.44, 0.59] | 0.166 | 0.135 | **0.166** | 0.008 | BL-D fall-back |
| PubMedQA ($N$=1k) | [0.90, 0.90] | 0.324 | 0.685 | **0.685** | 0.000 | CH-H deploy |
| Mixed 50:50 ($N$=2k) | [0.72, 0.76] | 0.244 | 0.411 | **0.411** | 0.005 | CH-H deploy |

Online adaptive thresholding—e.g., per-batch FR calibration with rolling estimates, or a learned threshold conditioned on query-distribution features—remains an open direction.

Table S41: Pilot latency overhead (median p50, single seed). The $\sim$3% average overhead reflects BM25 candidate-level scoring; deployment-safe.

| **Scenario** | **BL-D** | **Hybrid** | **Δ ms** | **Δ %** |
|---|---|---|---|---|
| NQ-only | 1670 | 1684 | +14 | +0.8% |
| PubMedQA-only | 1671 | 1757 | +86 | +5.1% |
| Mixed 50:50 | 1668 | 1724 | +56 | +3.4% |
| **Mean** | **1670** | **1722** | **+52** | **+3.1%** |

**Summary.** The pilot empirically confirms the two-layer diagnostic: Layer 1 escape is *necessary but not* *sufficient*; Layer 2 (task-format compatibility) gates downstream EM gain. The negative result in NQ-only is not a failure of the protocol but a successful prediction of a Layer 2 violation under a deliberately stress-tested query distribution. Paper main results replicate within $\pm$1pp, and latency overhead averages $+3.1\%$ (Table S41)—deployment-safe.

# M Anchor Symmetry and Budget-U Robustness

This appendix reports two robustness experiments added in this revision: (E2) verifying anchor symmetry by reversing the roles of `wiki_full` and `MedRAG_PubMed` (addressing the anchor-symmetry open item noted in Limitations); and (E3) verifying that Finding 1's FR=0.000 is not an artifact of Budget-F by re-running CH-D/CH-H under unconstrained budget. All conditions use Llama-3.1-8B-Instruct, GTE-large-en-v1.5, the Mixed benchmark ($N$=500, 50:50 NQ:PubMedQA), and three seeds $\{0, 1, 42\}$.

## M.1 E2: Anchor Symmetry under PubMed-Anchor

We reverse the anchor role: `MedRAG_PubMed` becomes the primary ($d_1^*$ source) and `wiki_full` is the secondary, with all other settings ($k$=5, Budget-F, GTE embedder) held identical to the wiki-anchor experiments of §6.

Table S42: E2 – Anchor symmetry verification (3-seed mean): the PubMed-anchor configuration reproduces Layer 1 (FR=0.000) of the wiki-anchor setting. Rows use BL-P (single-source PubMed) and the PubMed-anchor multi-source conditions (wiki secondary); wiki-anchor reference rows from Table 2.

| **Condition** | **Fusion** | **EM** | **NQ** | **FR** |
|---|---|---|---|---|
| *PubMed-anchor (this revision)* | | | | |
| BL-P | — | 0.336 | 0.024 | — |
| CH-D (PubMed-anchor) | rrf | 0.383 | 0.137 | **0.000** |
| CH-H (PubMed-anchor) | hybrid | 0.397 | 0.151 | 0.596 |
| *Wiki-anchor reference (paper main)* | | | | |
| BL-D | — | 0.243 | 0.163 | — |
| CH-D | rrf | 0.366 | 0.135 | **0.000** |
| CH-H | hybrid | 0.406 | 0.133 | 0.724 |

**Layer 1 is universal.** Dense RRF FR=0.000 is reproduced exactly under role reversal (Table S42): regardless of which corpus serves as anchor, Dense RRF mechanically suppresses the secondary at top-1. This confirms Layer 1 suppression is a structural property of the retrieval method, not of `wiki_full`'s specific scale or geometry.

**Layer 2 magnitude is anchor-direction-dependent.** Hybrid FR lift is smaller under PubMed-anchor (+0.596) than under wiki-anchor (+0.724), and the Hybrid EM lift over Dense is correspondingly smaller (+1.4pp vs. +4.0pp). We attribute this to lexical-density asymmetry: medical (PubMedQA) queries more reliably surface discriminative tokens in `MedRAG_PubMed` than general-knowledge (NQ) queries surface them in `wiki_full`, so BM25's lexical activation operates more strongly in the wiki-anchor direction. The qualitative two-layer pattern (Dense suppresses; Hybrid escapes; EM modestly improves) is preserved.

**NQ recovery under FR=0.** BL-P NQ EM=0.024 rises to the PubMed-anchor CH-D NQ EM=0.137 (+11.3pp) *despite* FR=0.000. This is the predicted Budget-F mechanism (§6.1, Finding 1): with $d_1^*$ suppressed, secondary passages still occupy positions 2–5 and contribute via context. The phenomenon is symmetric across anchor choices, providing additional evidence that retrieval-level suppression and context-level contribution are dissociable.

## M.2 E3: Budget-U Verification of Finding 1

To rule out Budget-F as the source of FR=0.000 for the content-heterogeneous setting (CH), we re-run the wiki-anchor CH-D/CH-H configurations under **Budget-U**: top-$k$ passages without per-source quotas. Table 5 already reports Budget-U for FH-D (formal heterogeneity, NQ baseline); this extends the test to CH (content heterogeneity, Mixed benchmark).

Table S43: E3 – Budget-U verification of Finding 1 (3-seed mean): FR=0.000 and the $+12.3$pp lower-rank EM contribution are reproduced without Budget-F. Reference Budget-F rows from Table 2.

| Condition | Fusion | EM | NQ | FR |
|---|---|---|---|---|
| *Budget-U (this revision)* | | | | |
| BL-D | — | 0.225 | 0.171 | — |
| CH-D | rrf | 0.348 | 0.136 | **0.000** |
| CH-H | hybrid | 0.404 | 0.156 | 0.715 |
| *Budget-F reference (paper main)* | | | | |
| BL-D | — | 0.243 | 0.163 | — |
| CH-D | rrf | 0.366 | 0.135 | **0.000** |
| CH-H | hybrid | 0.406 | 0.133 | 0.724 |

**FR is budget-invariant for CH.** CH-D FR remains 0.000 under unconstrained budget (Table S43), confirming the suppression is a retrieval-geometry property (§6.1, Finding 1), not an artifact of equal-allocation budget. Architecture invariance, previously demonstrated for FH-D in Table 5, now extends to CH.
**Lower-rank EM contribution is preserved.** $\Delta$EM(CH-D $-$ BL-D) = $+12.3$pp under Budget-U ($0.348 - 0.225 = 0.123$) matches the $+12.3$pp lower-rank contribution decomposed in §6.1, Finding 2. The Budget-F protocol therefore isolates the lower-rank contribution for measurement; it does not inflate it.
**Hybrid top-1 activation is preserved.** $\Delta$EM(CH-H $-$ CH-D) = $+5.6$pp under Budget-U ($0.404 - 0.348 = 0.056$) is slightly larger than the $+4.0$pp under Budget-F, with FR lift of comparable magnitude (0.715 vs. 0.724). The direction and significance of Hybrid top-1 activation are preserved across budget regimes.

## M.3 Joint Implication and Per-Seed Reproducibility

Together, E2 and E3 close two open robustness items: anchor symmetry (E2 closes the anchor-symmetry open item from Limitations) and Budget-F sensitivity of the content-heterogeneous case (E3 extends Table 5 from FH to CH). The two-layer diagnostic survives both perturbations: Layer 1 (FR=0 under Dense RRF) is universal; Layer 2 (Hybrid EM lift) is bounded but direction-preserving across anchor and budget regimes.

**Per-seed values for reproducibility.** *E2 (PubMed-anchor)*: BL-P EM = 0.334/0.340/0.334 (seeds 0/1/42); PubMed-anchor CH-D EM = 0.378/0.388/0.382, FR = 0/0/0; PubMed-anchor CH-H EM = 0.402/0.396/0.394, FR = 0.598/0.592/0.598. *E3 (Budget-U)*: BL-D EM = 0.210/0.244/0.222; CH-D EM = 0.346/0.346/0.352, FR = 0/0/0; CH-H EM = 0.406/0.402/0.404, FR = 0.720/0.712/0.714. Per-seed bootstrap CI95 widths are $\leq\pm0.044$ on $N$=500; cross-seed standard deviations are $\leq$0.018 for EM and $\leq$0.004 for FR, indicating stable estimates.

# N Supervised Routing Baseline: Linear Classifier from N=200 Labels

**Setup.** We train a logistic-regression classifier (sklearn defaults, $C$=1, max_iter=2000) on $N$=200 dev queries (100 NQ + 100 PubMedQA). Each query is encoded with GTE-large-en-v1.5 (the same encoder as our main pipeline), producing 1024-dimensional features. At inference, the classifier predicts $\in \{\text{NQ}, \text{PubMed}\}$

and routes each Mixed-test query to the corresponding single-source baseline (BL-D for NQ, BL-P for PubMed).

**Dev set (disjoint from test).** NQ dev queries are sampled from `nq_open` train (87,925 queries; disjoint from validation/test). PubMedQA dev queries are sampled from `pqa_artificial` (211,269 queries; disjoint from `pqa_labeled` which is the paper test set). Random seed=42.

**Sensitivity.** Dev self-accuracy is 1.0 at $C \in \{0.1, 1, 10, 100\}$ and 0.99 at $C$=0.01, confirming classifier robustness.

**Results on Mixed test ($N$=2,000, seed=42).** Routing accuracy is **0.986** (95% bootstrap CI [0.981, 0.991]; NQ partition 0.980, PubMed partition 0.992). Routed EM is **0.4065** (95% CI [0.385, 0.428])—within 0.05pp of CH-H (0.406) and 0.3pp of the per-query oracle (0.4095). On the same seed=42 partition: BL-D 0.247, BL-P 0.336.

**Interpretation.** CH-H matches the supervised router within 0.05pp while requiring no per-query labels at inference, and both approach the per-query oracle (0.4095). Routing accuracy 98.6% confirms NQ vs PubMedQA queries are near-linearly separable in GTE embeddings; the remaining 1.4% misroutes account for the 0.3pp gap from oracle. This establishes label-free CH-H as a competitive alternative to supervised routing under aggregate EM; evaluation on harder source-discrimination tasks (e.g., overlapping domains) is future work.

# O Candidate-Level BM25 as the Routing Mechanism: Per-Query Evidence

This appendix supplies the per-query evidence underlying Finding 2 (§6.1) that candidate-level Hybrid acts as a label-free query-conditional source router. We unpack three claims: (i) the structural preconditions for the FR=0 lock-in are met on every query in our Mixed benchmark; (ii) candidate-level BM25 over the 10-doc Dense pool produces no ties and assigns top-1 in a partition-conditional way; (iii) BM25's source assignment matches the final CH-H top-1 source assignment to within sampling noise, identifying BM25 as the proximate router.

**Preconditions for the per-source RRF tie hold universally.** On the Mixed seed=42 run ($N$=2,000; equivalent counts on seeds 0/1), we audit the structural conditions stated in §4.2 (*Mechanistic attribution and pathology scope*):

- Both sources return $k$=5 candidates on every query (no empty results from either `wiki_full` or `MedRAG_PubMed`: 2,000/2,000).
- Cross-source doc-id overlap is empty on every query (LOI=0 on 2,000/2,000); the corpora are disjoint at the document-id level.
- The top-1 and top-2 Dense-RRF scores are exactly tied at $1/(60+1) \approx 0.016393$ on every query (2,000/2,000); fused top-1 is decided by source-list ordering and is `wiki_full` on 2,000/2,000 (the wiki-anchor convention).

These three audits jointly verify that FR=0 under Dense RRF is the structural consequence of per-source-$k$ rank fusion described in §4.2, not a contingent property of the specific encoder or query set.

**Candidate-level BM25 breaks the tie partition-conditionally.** We re-audit the same $N$=2,000 run under the Hybrid fusion (§4.3). The number of queries with a top-1/top-2 score tie drops from 2,000 (Dense) to $264/2{,}000$ (Hybrid; the residual ties occur on queries where BM25 returns identical zero scores for the top candidates—see below). Top-1 source assignment is strongly partition-conditional: NQ queries select `wiki_full` top-1 in 777/1,000 (77.7%); PubMedQA queries select `MedRAG_PubMed` top-1 in $896/1{,}000$ (89.6%); aggregate `wiki_full` 881/2,000 (44.0%), `MedRAG_PubMed` 1,119/2,000 (56.0%). The pattern is consistent with BM25's bias toward lexical overlap: biomedical queries surface specialist vocabulary present in `MedRAG_PubMed` abstracts, general queries surface entity/event mentions in `wiki_full` articles.

**BM25 candidate-pool ranking matches CH-H top-1 source assignment.** We compare the candidate whose BM25 score is highest in the 10-doc pool ("BM25 top-1") to the Hybrid fusion's top-1. Aggregate BM25-top-1 source distribution: `wiki_full` 841/2,000 (42.0%), `MedRAG_PubMed` 1,159/2,000 (58.0%); CH-H top-1 source distribution: `wiki_full` 881/2,000 (44.0%), `MedRAG_PubMed` 1,119/2,000 (56.0%); the two distributions differ by $\leq$ 2pp per source, and the 264 residual ties (BM25 score 0 on both candidates) account for the small mismatch. Per-query, the BM25 top-1 and Hybrid top-1 coincide in source identity on the overwhelming majority of queries; the disagreements concentrate in queries where Dense-RRF's $1/(60+1)$ tie-contribution to wiki marginally outweighs a low BM25-rank gap.

**Implication.** These three audits jointly substantiate the mechanism stated in §6.1, Finding 2: candidate-level BM25's global ranking over the joint Dense pool is the proximate driver of Hybrid top-1 source assignment, and its partition-conditional behavior (NQ→wiki, PubMedQA→PubMed) is what realizes the label-free oracle-equivalent routing reported in §6.1. The mechanism is, in this sense, mechanistically transparent: a global lexical-density ranking over a small joint candidate pool serves as an implicit query-domain classifier without any per-query labels.

**Routing does not collapse on lexically ambiguous queries.** A natural concern is that the constructed NQ/PubMedQA mixture pairs two lexically well-separated populations, so the 0.83 routing accuracy might reflect only easy, self-announcing queries. We test this directly. For each query we compute a *lexical separability* score $\mathrm{sep}(q) = |\, b^{\max}_{\text{wiki}} - b^{\max}_{\text{pubmed}} \,|/(b^{\max}_{\text{wiki}} + b^{\max}_{\text{pubmed}}) \in [0,1]$, where $b^{\max}_{s}$ is the highest BM25 score among source $s$'s candidates in the joint pool; low sep means the two sources are near-indistinguishable lexically for that query. Binning the $N$=1,897 content-covered queries (seed 42) into separability quintiles, routing accuracy rises monotonically with separability—$0.654, 0.770, 0.868, 0.921, 0.958$ from the most ambiguous to the most separable quintile—but crucially, even the *most ambiguous* quintile ($\mathrm{sep} \leq 0.084$) routes at $0.654$, well above the $0.50$ chance floor. The routing signal thus degrades gracefully rather than collapsing: candidate-level BM25 recovers query-domain alignment even where the two corpora are lexically hard to tell apart, indicating the mechanism is not an artifact of maximal inter-partition separation.

**Breaking the tie is trivial; routing by query domain is not.** To separate *escaping FR=0* from *routing correctly*, we re-fuse the same Dense candidate pool under four alternative rules and measure routing accuracy (Table S44, 3-seed means, per-cell std $\leq$0.008). A random float tie-break, raw CombSUM, and per-source z-CombSUM all raise FR above 0—so escaping the lock-in is trivially achievable—yet all route at chance (0.50–0.53): raw CombSUM is scale-dominated (top-1 always `MedRAG_PubMed`), and neither random nor z-scored fusion carries a query-domain signal. Only candidate-level Hybrid routes by query domain (0.83; NQ→wiki 0.765, PubMedQA→PubMed 0.896) and thereby matches the supervised/oracle EM without labels. The non-trivial contribution is thus the accurate query-domain routing BM25's lexical signal provides, not the continuous score that breaks the tie. (We report routing accuracy, not EM, here: a chance tie-break can partly help EM by luck—breaking anchor lock-in on the PubMedQA half—so EM would overstate its value; routing accuracy isolates the substantive quantity.)

Table S44: Escaping FR=0 is trivial (many rules do it); query-domain routing is not (only candidate-level Hybrid attains it). Re-fusion of the identical Dense candidate pool, Mixed benchmark, 3-seed means.

| Fusion rule | FR | Routing acc. |
|---|---|---|
| RRF (CH-D) | 0.000 | 0.500 |
| + random tie-break | 0.499 | 0.500 |
| CombSUM (raw dense) | 1.000 | 0.500 |
| z-CombSUM (per-source $z$) | 0.549 | 0.517 |
| **Hybrid (Dense+BM25)** | **0.724** | **0.830** |

# P Score-Level Diagnosis of Dense Partition Signal: z-CombSUM as a Measurement Tool

Appendix O establishes candidate-level Hybrid as the *deployment* mechanism that realizes label-free routing. This appendix isolates the complementary *measurement* question: *does the Dense embedding score itself contain a partition-discriminative signal that per-source rank fusion discards?* We answer in the affirmative via a label-free score-level diagnostic, per-source z-score CombSUM (henceforth *z-CombSUM*), and show that the diagnostic correctly returns *no* signal under same-domain heterogeneity—validating the measurement as a property of the data rather than an artifact of the fusion rule.

**Setup.** For each query, we collect the per-source top-$k$=5 candidates produced by Dense retrieval (the same pool consumed by RRF and Hybrid). We compute per-source mean and standard deviation of raw Dense scores over the top-$k$ pool, standardize each candidate's score by its source's $(\mu, \sigma)$, and rerank the joint pool by standardized score (a per-source z-score variant of CombSUM [Fox and Shaw, 1994]). The top-1 source under this reranking is reported as the z-CombSUM routing decision. No labels, classifiers, or BM25 are used; the inputs are exactly the Dense scores already saved in the retrieval logs.

**Score-scale heterogeneity is structural.** The top-$k$ pool exposes a $\sim 23\times$ scale gap between sources under our main cross-domain (CH) configuration (`wiki_full`: mean 0.776, range $[0.59, 0.95]$ on 10,000 candidates; `MedRAG_PubMed`: mean 17.84, range $[11.47, 22.90]$). The gap is consistent across seeds (Table S45: 22.97/22.98/23.00$\times$) and the gap reproduces in the same-domain configuration with a different medical corpus (`MedRAG_Wikipedia`: 24.16$\times$). We interpret this as the *empirical manifestation of score incomparability* (the mechanism-(ii) regime named in §4.2): when corpora are indexed independently—even nominally with the same encoder—the absolute Dense-score magnitudes are not commensurable, and any fusion rule that ignores scale (RRF) discards the signal carried by within-source distributional position.

**z-CombSUM recovers partition-conditional routing without BM25 or labels.** On the cross-domain configuration (wiki + PubMed), z-CombSUM routes NQ queries to `wiki_full` in 94.93% of cases and PubMedQA queries to `MedRAG_PubMed` in 97.50% of cases, averaged over seeds $\{0, 1, 42\}$ (Table S45; cross-seed std $\leq 0.81$pp). The same per-query Dense scores that RRF locks into 100% wiki on both partitions (Appendix O) carry near-perfect partition information once their per-source scale is removed. This establishes that the per-source-$k$ RRF lock-in identified in §4.2 is fusion-level, not signal-level: the Dense encoder already separates NQ-relevant from PubMedQA-relevant documents in its (per-source) score distribution.

| Seed | Scale ratio | NQ→wiki (z-CombSUM) | PMQA→PMed (z-CombSUM) | RRF top-1 (both partitions) |
|---|---|---|---|---|
| 0 | 22.97× | 94.9% | 97.7% | 100% wiki |
| 1 | 22.98× | 95.0% | 97.4% | 100% wiki |
| 42 | 23.00× | 94.9% | 97.4% | 100% wiki |
| mean | 22.98× | 94.93% | 97.50% | 100% wiki |
| std | 0.02× | 0.06pp | 0.17pp | — |

Table S45: Cross-domain (wiki + PubMed) z-CombSUM routing across seeds. Routing accuracy is seed-invariant; RRF lock-in is universal. $N$=2,000 per seed (1,000 NQ + 1,000 PubMedQA).

| Seed | Scale ratio | NQ→wiki | PMQA→wiki |
|---|---|---|---|
| 0 | 24.16× | 50.9% | 48.7% |
| 1 | 24.16× | 52.5% | 48.8% |
| 42 | 24.16× | 51.5% | 48.9% |
| mean | 24.16× | 51.63% | 48.80% |

Table S46: Same-domain (wiki + MedRAG_Wikipedia) z-CombSUM routing across seeds. Same-domain pairing produces $\sim$ 50:50 on both partitions, confirming the diagnostic returns no signal where none exists. RRF (not shown) is 100% wiki on both partitions, identically to the cross-domain case.

**Same-domain control confirms the diagnostic is signal-driven, not artifact-driven.** The same-domain control pairs `wiki_full` with `MedRAG_Wikipedia`—two wiki-derived corpora with low content heterogeneity. If z-CombSUM's cross-domain routing were an artifact of per-source standardization (e.g., always favoring the higher-variance source), the same-domain control would inherit a consistent bias. Instead, z-CombSUM returns $\approx$ 50:50 on both NQ and PubMedQA partitions across all three seeds (Table S46: `wiki_full` share 51.6% $\pm$ 0.81 on NQ, 48.8% $\pm$ 0.10 on PubMedQA). The diagnostic correctly reports the absence of partition-discriminative content, aligning with the *content-heterogeneity* axis of our framework (§3.2): same-domain pairings have nothing to route between.

**Relationship to candidate-level Hybrid.** z-CombSUM and Hybrid resolve the same RRF tie via different routes. z-CombSUM is a *measurement* tool: it diagnoses whether a partition signal exists in the Dense score distribution, and quantifies the magnitude lost to rank conversion. Hybrid is the *deployment* instantiation: candidate-level BM25 reranking is one effective realization that translates the latent partition signal into downstream gains (mixed EM +4.0pp aggregate, PubMedQA EM up to +8.3pp on Llama; Table 2 and Figure S1). The two operate at different layers and are complementary rather than competing: Hybrid's downstream EM lift is empirically established (Tables 1, 2); z-CombSUM's downstream effect is bounded by Budget-F's per-source quota, which already exposes both sources to the generator and limits the margin top-1 reranking can produce in EM. We therefore position z-CombSUM as a diagnostic axis within the measurement protocol rather than a substitute for the deployment fusion.

**Reproducibility tables.**

**Caveats.** (a) Per-source $(\mu, \sigma)$ are estimated from the top-$k$ pool rather than the full corpus; this approximates the corpus distribution with the retrieved high-score tail and may slightly attenuate within-source spread. The diagnostic still produces the qualitative split (cross-domain partition-conditional, same-domain null), supporting robustness, but full-corpus standardization would refine the magnitude estimate. (b) z-CombSUM is evaluated at the top-1 source-routing layer; we do not claim it improves downstream EM. Budget-F's per-source token quota exposes both sources to the generator on every query, capping the EM margin that

| Fusion | FR | Routing acc. |
|---|---|---|
| raw CombSUM | 1.000 | 0.477 (chance) |
| z-CombSUM | 0.549 | 0.517 (chance) |
| min-max CombSUM | **0.000** | 0.523 |
| convex interp. $\alpha$=0.7 (dense-dominant) | 1.000 | 0.477 |
| convex interp. $\alpha$=0.5 | 0.999 | 0.479 |
| convex interp. $\alpha$=0.3 (BM25-dominant) | 0.937 | 0.704 |
| candidate-level Hybrid (ours) | 0.706 | **0.835** |

Table S47: Standard score-level fusion baselines, re-fused offline on the identical cached candidate pool (seed 42, $N$=1,897). Min-max CombSUM *re-creates* the tie-lock exactly—every source's rank-1 normalizes to 1.0, so FR=0 with the anchor winning by list order. CombMNZ is omitted as a row: with disjoint corpora every document appears in exactly one source list (hit count = 1), so CombMNZ coincides with CombSUM analytically. The best convex interpolation ($\alpha$=0.3) routes at 0.704, 13pp below Hybrid; dense-dominant settings collapse to chance. FR/routing for Hybrid on this $N$=1,897 subset (0.706/0.835) match the full-set values (0.724/0.83).

top-1 reranking can produce. The intended contribution of z-CombSUM here is diagnostic—establishing that the partition signal exists in Dense scores—not deployable improvement. (c) The cross-domain result promotes the mechanism-(ii) score-incomparability claim from a stated mechanism in §4.2 to an empirically measured property; the same-domain control confirms the measurement responds to content heterogeneity rather than producing a constant signal.

**Implication.** The two-layer mechanism articulated in the main text—rank-level pathology (Appendix O) and content-routing remedy (§6.1, Finding 2)—is bracketed by a score-level measurement layer. Dense scores carry a partition-discriminative signal under content heterogeneity that per-source-$k$ rank fusion discards (mechanism (ii)); candidate-level BM25 reranking restores routing in a deployment-ready form (mechanism (iii) of §4.2); the same-domain control verifies that the measurement is signal-driven. Together, the three perspectives (rank-tie, score-incomparability, content-routing) form a single internally consistent account of when and why multi-source RAG helps.

## P.1 Standard Score-Level Fusion Baselines: Offline Re-Fusion

Beyond raw and z-scored CombSUM, we re-fuse the identical cached joint candidate pool (Medical Mixed, seed 42) under the remaining standard score-level fusion families: per-source min-max CombSUM, CombMNZ, and convex Dense–BM25 interpolation $\alpha \cdot \mathrm{minmax}(\text{dense}) + (1-\alpha) \cdot \mathrm{minmax}(\text{BM25})$ (cf. single-corpus hybrid interpolation). All variants operate on the same per-source dense scores and joint-pool BM25 scores as candidate-level Hybrid; no retrieval or generation is re-run. Queries with full candidate-content coverage in the cache are used ($N$=1,897/2,000); as validation, our offline re-fusion reproduces the cached CH-H top-1 exactly (100% of queries).

The pattern is uniform: score-level normalization either preserves the tie (min-max re-ties every rank-1 at 1.0), inherits cross-source incomparability (raw, z, dense-dominant interpolation route at chance), or—when BM25 dominates the mixture—approaches but does not reach the rank-based two-stage RRF (0.704 vs. 0.835). Candidate-level Hybrid's advantage is thus not an artifact of comparing only against weak tie-breaks: it holds against the standard score-level fusion families under identical retrieval.

# Q Reproducibility Details

This appendix consolidates implementation specifics referenced in §5.4 for direct re-execution.

## Q.1 Prompt Templates (verbatim)

All experiments with Llama-3.1-8B-Instruct use the following templates, with `do_sample=False`, `num_beams=1`, `temperature=0.0`, `max_new_tokens=64`, `max_input=2048`.

### Answer-only (NQ, HotpotQA).

```
Answer the question based on the following passages.

{passages}

Question: {question}
Answer:
```

### PubMedQA (yes/no/maybe).

```
Based on the following passages, answer the biomedical
question with yes, no, or maybe.
Respond with ONLY one of: yes, no, maybe.

{passages}

Question: {question}
Answer:
```

### FEVER (3-class classification).

```
Based on the following passages, determine whether
the claim is SUPPORTS, REFUTES, or NOT ENOUGH INFO.
Respond with exactly one of: SUPPORTS, REFUTES,
NOT ENOUGH INFO.

{passages}

Claim: {claim}
Label:
```

**Passage formatter.** Each retrieved document is rendered as `[i] (source_name) title \ncontent` and separated by a blank line.

**Answer parsing.** PubMedQA: first occurrence of `yes`/`no`/`maybe` (case-insensitive, after stripping punctuation); fallback `maybe`. FEVER: priority match for `NOT ENOUGH INFO`, then `SUPPORTS`/`REFUTES`; fallback `NOT ENOUGH INFO`. NQ/HotpotQA: first non-empty line, stripped.

## Q.2 Vector Index Configuration

Milvus collections use DISKANN with inner-product (IP) metric:

```
collection.create_index(
    field_name="vector",
    index_params={"index_type": "DISKANN",
                  "metric_type": "IP"},
)
```

Default Milvus DISKANN parameters are used; embeddings are unit-normalized so IP equals cosine similarity. Embedding model: GTE-large-en-v1.5 (dim=1024).

## Q.3 BM25 Configuration

Candidate-level Hybrid: `rank_bm25` ($k_1$=1.5, $b$=0.75), scoring over Dense top-$k$ documents. Full-corpus BM25 ablation (§6.2) uses `bm25s` with identical hyperparameters on the full `MedRAG_PubMed` corpus (23.9M chunks; index size ∼15GB).

## Q.4 Dataset Splits

**PubMedQA**: `qiaojin/PubMedQA`, config `pqa_labeled`, split `train` (the 1,000-sample expert-labeled subset); first $N$ items used after deterministic load (no shuffle). **NQ**: standard validation split, subsampled to $N$=2,000 per seed (seeds 0, 1, 42; seed-deterministic subsample). **HotpotQA, FEVER, CUAD, FiQA-2018, TAT-QA**: standard public splits as released.

## Q.5 Seeds and Stochasticity

Three random seeds (0, 1, 42) control NQ subsample selection and shuffle-dependent operations. Llama generation is greedy (`do_sample=False`), so generator output is deterministic given inputs.

## Q.6 Compute Resources

All experiments run on NVIDIA L40S GPUs (48 GB GDDR6, 350 W TDP); the Llama-3.1-8B-Instruct pipeline fits a single GPU per experiment without distributed inference. Total compute is approximately 200 GPU-hours, distributed as: ∼10h vector indexing (one-time across six corpora), ∼10h Phase 1 baselines (4 datasets × 3 seeds), ∼50h Phase 2 main results (9 conditions × 3 seeds × $N$=2,000; Table 2), ∼60h four-LLM generalization (Table S2), ∼40h cross-domain validation (App G), ∼10h ablations (Table 5), and ∼20h pilot (App L) and supplementary appendices. Peak GPU memory is ∼22 GB for Llama-3.1-8B-Instruct inference (single-GPU, BF16); the Milvus DISKANN index is disk-resident (∼60 GB combined corpora storage). Estimated emissions: ∼28 kg$CO_2$eq (ML $CO_2$ Impact calculator).

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